The power of population density in entrepreneurial ecosystems: attracting business to the place[1]

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Abstract: Business venturing explained with a theory of entrepreneurial ecosystems has traditionally focused on knowledge transfers between the most productive and innovative firms, typically relying on agglomeration mechanisms like matching, learning, and sharing. We develop a social dimension of entrepreneurial ecosystems by adding population density to a set of factors of carrying capacity. We argue that population density externalities complement these mechanisms with attracting, which is necessary to fully understand the theory of firms' births. Our study explores the existence of the hierarchical causal mechanism that occurs when 2nd line firms (non-innovative, less-productive retail and service businesses) and human settlement create an environment that attracts 1st line firms (the most innovative and productive businesses). Using an urban and peripheral context, we demonstrate that population density and business agglomeration jointly impact firms' placement in various sectors. Specifically, we reveal the functionally and spatially hierarchical impact of density and agglomeration externalities on business location decisions, where peripherality, local specialization, diversity, competition, and urbanization externalities play significant roles.

JEL Code: L25, R32, D62, R12

Keywords: entrepreneurial ecosystem, carrying capacity, economies of density; attracting mechanism; population density; hierarchical agglomeration; convenience; business venturing

Executive summary: This paper explores the relationship between population density and entrepreneurial ecosystems. The paper argues that population density is a crucial determinant of the carrying capacity of an entrepreneurial ecosystem and introduces the concept of population density externalities, which operate through an attraction mechanism. The study hypothesizes that there are two parallel channels of business attraction: one through agglomeration externalities and the other through population density externalities. The paper presents empirical evidence supporting these hypotheses, utilizing mediation, binary-choice, and elasticity models.

The research findings have several implications for previous studies on business venturing and entrepreneurial ecosystems. The paper reveals that population density plays a significant role in firms' location decisions and should be considered alongside agglomeration economies. The attraction mechanism operates on different spatial levels and applies to various business sectors. The study identifies a stable level of elasticity between business agglomeration and residential population, demonstrating that

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increasing population density leads to increased business activity. The results highlight the importance of incorporating population impact into the understanding of entrepreneurial ecosystems and suggest that population density should not be overlooked in enterprise venturing studies.

Overall, this paper contributes to the literature on entrepreneurial ecosystems by enriching the social dimension and providing empirical evidence for the role of population density. It emphasizes the need to consider both agglomeration economics and the effects related to the spatial location of firms. The findings open up new avenues for research in entrepreneurship, business location, and business demography, and underscore the importance of accounting for population density in understanding the dynamics of entrepreneurial ecosystems.

Highlights:

- The theory of entrepreneurial ecosystems includes population density as a determinant of carrying capacity
- Population density is the cause of business agglomeration
- Economies of density and the economies of agglomeration jointly matter for business location decisions
- Duranton and Puga's (2004) agglomeration mechanisms of sharing, learning, and matching in business locations must be supplemented by the attraction mechanism of density
- The attraction mechanism can be explained by economic relations and phenomena of convenience and liveable city
- Location determinants are mediated – the direct influence of population and the indirect influence of business agglomeration is observed
- Business agglomeration is hierarchical – functionally due to links between 1st and 2nd line firms, and spatially due to localised effects of urban settlement
- The elasticity of business agglomeration with regard to population density is 0.76 and it evidences localised effects due to distance to cities of different size

Introduction

Entrepreneurship theory since long tried to explain how and why ventures appear (Packard & Burnham, 2021) and one of the promising streams is the theory of entrepreneurial ecosystem (EE). It operates in three dimensions: economic, technological, and societal (Audretsch et al., 2019), which interact. Despite many studies in this field (e.g. Cavallo et al., 2019; Cao & Shi, 2021), the linkages between entrepreneurship dynamics and population density remain neglected, which is the consequence of generally lower attention paid to the societal dimension of EE (Audretsch et al., 2019). In fact, population density is a crucial determinant of the carrying capacity of EE, but missing until now in definitions. Abootorabi et al. (2021) define carrying capacity as the "number of organisms that can be sustained in a particular ecosystem" (understood as a number of ventures), and make it dependent on a number of firms in EE, available resources and demand for those resources. EE are analysed within spatially limited borders, and locally-diversified models explain entrepreneurship better than globally-fitted approaches (Iacobucci & Perugini, 2021; Muñoz et al., 2022). Those local ecosystems may differ significantly in all aspects, but what is already known, the geo-proximity of stakeholders as well as a higher concentration of resources promote entrepreneur behaviour (Abootorabi et al., 2021).

Recent literature in business venturing has largely focused on the most attractive firms which are knowledge-based, patenting, exporting and intensively growing (we call them 1st line firms). This has focused the analysis on the relations between these firms, including agglomeration externalities, knowledge flows, proximity and spillovers (e.g. Eriksson, 2011, Iammarino & McCann, 2013). Comprehensive theoretical justification of this micro-world was started by Marshall (1890), developed within New Economic Geography and well-defined with mechanisms of matching, learning and sharing by Duranton & Puga (2004). This zoomed analytical perspective, even if explaining in detail this most prominent layer of economic activity is missing broader context in two aspects: first, the remaining part of firms which are mostly local retail, financial and personal services providing the economic base and facilities (we call them 2nd line firms, see Appendix 7), and secondly, human settlement.

The fundamental question in business venturing is how successful ventures are created in relation to people. One of the convincing theories, among many others, is the theory of empathy (Packard & Burnham, 2021), in which firms are created by people for people. Empathy, which is an understanding of the real needs of customers, determines entrepreneurial performance and survival. It is close to the theory of co-creative entrepreneurship (Karami & Read, 2021), which underlines the externalities of joint actions of stakeholders – benefits to all participants from acting together. Even if this theory assumes (implicitly) realized and aware collaboration, it seems that demand-supply-expressed goals can be co-creatively achieved in a crowded entrepreneurial ecosystem without communication. This is because human and business populations constitute the entrepreneurial ecosystem of a given place. The interactions of stakeholders in this local EE generate externalities which enrich the system and increase its carrying capacity. Habitat, being an environment of community, involves interactions of stakeholders, and the human population is its integral element. The most common externalities generated by firms located in the same local entrepreneurial ecosystem are matching, learning and sharing mechanisms, which cover flows of workers, knowledge, ideas, entrepreneurship spirit, organisation culture etc (e.g. Capello, 2002). What has not been discussed in literature until now, is that population density creates externalities of attraction – and works by attracting others to the place. This phenomenon can be explained by economic factors, convenience aspects and spatial organisation of cities (see section 2). We claim that the attracting power of economies of population density enables establishing 2nd line firms providing daily-life infrastructure, which in consequence, may create an attractive environment for 1st line businesses.

This paper develops the business venturing theory by adding a population density to a set of determinants of the entrepreneurial ecosystem (EE) and by including population density externalities, operating through an attraction mechanism, in the carrying capacity of EE. We hypothesise that there exist two parallel channels of business attraction in the entrepreneurial ecosystem: the well-known one through business agglomeration externalities and the one described here through population density externalities. They are supposed to work hierarchically – population density, as the primary cause, attracts all types of ventures to the place - the most productive ones (1st line) and those providing the everyday-life infrastructure (2nd line). Additionally, 2nd line firms serve as a mediator – booster of 1st line firms' emergence. This agglomeration hierarchy has functional and spatial dimensions – as business agglomeration elasticity depends on population density and relative location to cities of different sizes.

We empirically evidence the existence of these mechanisms. We estimate three models: the mediation model, the binary-choice model and the elasticity model to evidence these phenomena. The mediation model shows causal hierarchical relationships between population, 1st and 2nd line firms. The binary-choice model provides a complete view of the determinants of a business location – for the first time in the literature embracing the effects of the economies of density. This model is estimated twofold: with probit (linear econometric model) and random-forest (non-linear supervised machine learning technique) – which is a novel application in business location research. The elasticity model provides evidence of the strength of the relations between population density and business agglomeration.

The remainder of the paper is as follows: section 1 develops the concept of entrepreneurial ecosystem by introducing the role of population density in carrying capacity and explaining its externalities by the mechanism of attraction. Economies of density are considered complementary to economies of agglomeration and considered interdisciplinary, using economic factors, convenience approach and spatial organisation design. Section 2 reviews and proposes methodological approaches to study hierarchical agglomeration, population density externalities and agglomeration elasticity in intra-urban and extra-urban entrepreneurial ecosystems. Section 3 presents an empirical study on the entire population of 1 million of 1st and 2nd line firms in the NUTS1 Mazovian region (Poland, with Warsaw capital city) including individual point data on the 5.4 million population based on 1 km x 1 km grid data. It shows that population density is an essential element of the business venturing model. Section 4 discusses the implications of population density externalities and presents conclusions.

1. Attraction mechanism of population density economies

Literature on firms' births can be split into two streams: the general stream which discusses universal mechanisms, and the spatial stream, which sees business entities in a localised context (Roundy & Fayard, 2019). The theory of entrepreneurial ecosystems often assumes that the agglomeration of firms and resulting agglomeration economies are the major factors attracting new firms to the area (Audretsch et al., 2019). Agglomeration economies, understood as benefits to firms from co-location/clustering to other firms, were widely explained by micro-foundation mechanisms of matching, sharing and learning (Duranton & Puga, 2004) and studied deeply (e.g. Behrens et al., 2014). Those benefits can be understood as a decline in average cost, financial and technological externalities, increasing returns to scale in production, first-mover advantages, and regional specialisation (Anas et al., 1998).

Interactions between stakeholders of the local entrepreneurial ecosystem generate externalities that increase its carrying capacity. Important stakeholders, beyond firms, are inhabitants of the given EE. Population density generates economies of density understood as benefits to firms from co-location to people. Below we show that the economies of population density can be explained well by the mechanism of attracting, which finds its roots in social sciences. This mechanism assumes that high population density attracts 2nd line retail and service firms, creating an attractive urban environment, that consequently attracts the most productive 1st line firms.

Explanation of this phenomenon can be purely economic, as well as broader, social and spatial. From an economic perspective, high population density is associated with a large local customer base, so the mass of customers is nearby. Therefore, a diversity of customer preferences together with a greater chance of satisfying diverse tastes, and low interaction costs in terms of distance and time, make the attraction mechanism work. From a social perspective, which is intertwined with economic issues, it is explained by the higher liveability and convenience of a given place. From a spatial perspective, diverse relative locations matter for the strength of the attraction mechanism that generates spatial heterogeneity. These mechanisms are discussed in detail in the following section.

1.1 Economic factors

The urban market has long been perceived as an attractive environment for business. This is mainly due to agglomeration economies, which are linked to external economies of scale (due to co-location) – external to the firm, but internal to the region, and typically divided into economies of localisation (due to neighbourhood of the-same-sector business) and urbanisation (due to neighbourhood of diverse businesses). Those effects are activated by mechanisms of sharing, learning and matching in relation to knowledge and workers. It is usually argued that economies of agglomeration in densely populated areas increase labour and firms' productivity, stimulate innovations and make consumption cheaper (i.e. by lowering transport/travel costs) (Duranton & Puga, 2020, McCann & van Oort, 2019).

However, very different economic mechanisms come into play when analysing "desert urban area", with population and no settled business. Porter (1997), in his strategy of rebuilding the American inner cities, starts with a pre-condition of development – a high density of population. Its role is to generate local demand to be covered by 2nd line firms, located nearby, providing the economic base and facilities, and increasing saturation in retailing, financial and personal services. This, combined with safety, transportation hubs, a strategic location to reach customers efficiently and quickly and local community involvement in generating economic stimulus, becomes an attracting factor for 1st line firms.

Urban growth achieved by local consumption is contradicted to urban growth based on export. Increasing local consumption and developing small retail businesses evidently requires a high-density of population. The role of a superior local intra-urban consumption-base in attracting high-value businesses and skilled workers (Markusen & Schrock, 2009) is mentioned together with the three other factors which push forward urban growth. These are i) import substitution with locally-produced goods and services, ii) creating more jobs locally in labour-intensive service sectors, iii) creating an attractive environment for local innovation, which can prospectively grow in scope and spatial range. The superiority of the consumption-base is understood as the number and variety of available goods and services, including high-quality brands and their convenient availability in time and space. Urban consumption can be seen explicitly economically as shopping and the use of facilities and services, but it can also be considered more broadly.

Glaeser et al. (2001) assume that 'demand for city' and 'demand for density' appear when people want to live close together. This is when urban inhabitants find spatial equilibrium through the optimisation process, which balances urban rent premium (city's impact on housing prices) with urban productivity premium (city's impact on business productivity and wages) and urban amenity premium (city's impact on life quality). Glaeser's determinants of the urban productivity premium lie in transportation costs and sharing, learning and matching mechanisms, while urban amenity premium refers directly to the concept of urban liveability, where daily life convenience is supplemented with culture and sports access and even the marriage market. All those urban premiums depend indirectly on population density within the city.

The existence and survival of firms, especially local ones, often depends on the customer base in the neighbourhood – its quality and quantity (Stam & van de Ven, 2021). In the Hoteling model, the density of stores increases with the density of the population. Empirical studies confirm this mechanism, however, they add a wealth factor. Wealthier urban districts (with the same population density as poor ones) enjoy higher density and variety of retailing and services (Meltzer & Schuetz, 2012), as the preferences of richer inhabitants are more heterogeneous (Waldfogel, 2008). In Lancaster's view (1990), demand for variety may have two sources: the taste for diversity of individuals, or diversity of tastes of individuals. This is confirmed by empirical studies: denser areas enjoy higher business diversity due to a larger customer base, while more affluent and denser districts have this effect multiplied due to inherited preferences of the rich for diversity. This is in line with "consumer city" (Glaeser, 2001), further related to "amenity value" (Giuliano et al., 2019) and confirmed with empirical studies (Tavassoli et al., 2016).

The dense urban environment should be analysed from two different perspectives: social and business, which presumably interact. In the social one, this environment consists of local inhabitants, local workers and commuters only passing by. These people can face urban density externalities: positive ones such as variety, convenience, and quick interactions, but also negative ones such as crowding, pollution, etc. Their attitude towards dense areas results from inherited preferences and utility from urban density externalities and can vary from negative to positive. From a business perspective, "decisions of firms" are taken by people who are the decision-makers in business. Location decisions they make in business may be purely impersonal, but the smaller the firm, the higher the chance that they are related to the personal preferences of decision-makers and based on the information gained in the urban environment. Then, the location of the firm becomes dependent on density, through not a business but a private channel.

1.2 Convenience

Recent literature appreciates and analyses the convenience of everyday life, often linking it to the concept of a liveable city. Convenience, being an element of life quality, can be defined as the density and diversity of easily accessible urban facilities (Zhong et al., 2020; Marcus, 2010), or broader within 5D theory of build environment by density, diversity, design, distance to transit and destination accessibility (Ewing & Cervero, 2010), while liveable city concept refers to a relationship between people and cities in terms of all people's living needs (Ye et al., 2020).

The concept of 15 minutes city (e.g. Allam et al., 2022) is to design lively urban areas – a mixture of residential, entertainment and commercial elements in nearby, with decentralised "production and consumption hubs", where densification refers to both population and service provision. Within this kind of urban structure, people and 2nd line firms create an attractive environment for 1st line firms. It is clear that a high-growth company employing specialised workers will prefer to locate next to similar companies (Duranton & Puga, 2004). However, given a guaranteed neighbourhood of a similar company, it will prefer an urban area full of amenities to a peripheral or desert location. On the other hand, mono-functional business districts are losing their attractiveness (Smętkowski et al., 2021), as without permanent residents they offer only limited urban facilities. This phenomenon reveals the existence of economies of density, which appear next to economies of agglomeration.

The urban facilities consist of food services (including restaurants, bars, coffee shops, fast foods etc.), convenient shopping (including groceries, bakeries, small shops etc.), financial services (such as ATMs, banks, insurance etc.), leisure and entertainment (as parks, cinemas, theatres, gym, art, etc.), beauty and health services (as hairdressers, doctors, daily spa, pharmacies, etc.), education (as kindergartens, schools, foreign language courses etc.), everyday life services (as laundries, mobile services, cobblers, etc.) and transport services (as bus stops, subways, shared bikes and scooters etc.). They are expected to be available within 5-20 minutes of walking distance from residential areas and work places to guarantee convenience in a liveable city (e.g. Wang et al., 2021). The first studies confirm that urban amenities stimulate creativity better than simple firm agglomeration (Wu et al., 2022 for Shanghai).

The population density is an attracting factor for 2nd line firms, who are responsible for creating urban facilities. The population density accounts for the existence of 2nd line firms, while their diversity increases with people's proximity. This phenomenon is treated as a basis principle in the literature on urban design. Jacobs & Appleyard (1987) underline that urban life requires a minimum mass of people interacting close to each other[2]. However, density without liveability may result in slums rather than vibrant cities (see Porter, 1997 on inner-city problems[3]). Therefore a mixture of living, working, shopping, and recreation urban functions must co-exist together, not within a single building, but rather using a variety of public places. Interaction of people that happens in the urban space, mostly through the availability of facilities within walking distance, is considered the main advantage of high-density inner-urban living (Buys & Miller, 2011). Therefore, urban density provokes many studies on quality of life, referring to sustainability, liveability, neighbourhood satisfaction etc (Howley et al., 2009; Bramley et al., 2009).

Population density through the concept of convenience and a liveable city can explain the location of firms. Regional science theory already decades ago studied convenience. Following Ullmann (1962), "*the problem remains to design cities to take advantage of scale economies and the other advantages of concentration, and at the same time to provide optimum liveability*". Also, Hägerstrand (1970) half-century ago considered the convenience of everyday life. His informal model discusses bundles of daily activities that happen in time and space. He shows that a liveable city, where home, work, facilities etc., are close to each other, allows more activities (bundles) to be performed within the limited time of the day and is the desirable spatial organisation of life, supporting the work-life balance. At the same time, long commuting, which may cause inaccessibility of services (due to distance or opening time), is perceived as an unattractive necessity due to the financial constraints of residents. Even if technological progress and digitalisation of daily life services allow for more flexible management of activities, still many activities need in-person presence. Hägerstrand (1970) implicitly apprises the high agglomeration of 2nd line firms, which follow the population density. This liveable city is treated as an attractive environment for well-paid workers, which implies that the most attractive firms should locate there. Facilities create neighbourhood quality, which was classified by Hägerstrand (1970) as survival, comfort or satisfaction. This quality impacts short-run daily activities, cultural participation or access to medical care but also, in the long-term, generates educational paths for children. That makes the logical chain in which population density attracts 2nd line firms, and they consequently attract 1st line firms.

Neo-classical economists are troubled by the concept of convenience as it is not easy to operationalise it and might underpin non-rational decisions. Measurement of convenience is possible mainly from the supply-side – by knowing if the service is well-placed, with convenient opening hours, delivered reliably, punctually and comfortably (Anderson et al., 2013). However, the demand side for convenience is less clear. Even if economists assume daily-life decisions to be a logical reason or information-based

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[2]Jacobs & Appleyard (1987) specify the minimum net density to support urban life as 30-60 people per acre of land, and maximum around 200 persons per acre.

[3]Missing liveability in densely populated inner-cities of American cities results often from social problems of inhabitants as high crime rates, low education, low interest in regular work.

decision-making process, they would often appear as a normative-affective decision-making model, drawn on emotions, values, normative beliefs, and habits (Amitai, 1988). An increasing body of literature shows that convenience is a prerequisite for choice, even if final decisions are based on affective-emotional reasons, preferences, lifestyles, flexibility, convenience etc. (Buys & Miller, 2011; Kenyon & Lyons, 2003; Anable & Gatersleben, 2005).

1.3 Spatial organisation of regions and cities

Business locates not only in the centres of crowded cities but also in peripheral, sparsely populated areas. Understanding the spatial organisation of regional coverage with population and firms is of high interest. However, one poorly explained element of business location is the elasticity of business agglomeration in relation to population density. To our knowledge, this concept has never been studied before. In terms of methodological approach, the problem can be compared to the elasticity of wages in relation to business density (Andersson et al., 2016) or elasticity of productivity (Maré & Graham, 2013) – those studies confirm the significant agglomeration elasticity, around 0.066 for productivity and 0.03 for wages. However, existing studies focus on intra-urban differences (Andersson et al., 2016) and do not consider extra-urban regional scale or do not consider location as an important factor (Maré & Graham, 2013) or operate on urban aggregates (Graham, 2007; Melo et al., 2009). Therefore, two important questions arise: a) how business agglomeration is related to population density and if the attraction mechanism is working, b) is this elasticity varying over space, especially due to the relative location (e.g. distance to core, regional or small cities). This is crucial to understand the spatial hierarchy of agglomeration and density externalities on the micro-level, jointly in the intra-urban micro-scale and intra-regional (extra-urban) macro-scale. Elasticity ε is to reveal the fixed parity of business-inhabitants over the territory. One can expect there is local variation in (global) elasticity, which reveals the localised externalities. Elasticity $\varepsilon<1$ prevents extreme overcrowding of land units as doubling the population density would cause a less-than-proportional increase in business units. Elasticity $\varepsilon>1$ indicates a high attractiveness of the location, higher for firms than for inhabitants. This is an important question, how the regional diversity of population density translates into the hot and cold spots of business agglomeration.

Economies of agglomeration and density often observed in urban areas, are important drivers of venturing location decisions in entrepreneurial ecosystems and can be studied in terms of their efficiency (Audretsch et al., 2019). Even if regional economists see them as belonging to the individual firm, urban planners already link them with the city and divide the efficiency of agglomeration economies into comprehensive and technical ones (Yao et al., 2022). Comprehensive efficiency measures whether a city operates at optimal production efficiency with respect to input factors such as land, labour, and capital. It can be increased by a larger market size, sharing labour pool and infrastructure, shorter travel time and saving resources. Technical efficiency compares cities by scale and can be improved via knowledge spill-overs channels and supply-chain integration. Population density plays a special role there. High population density is a positive factor in increasing market size, saving resources, sharing the labour pool and knowledge spill-overs. However, it may also be a negative factor by increasing congestion and crime, which decrease agglomeration economies' efficiency. The population density in this setting is one of the fundamental factors, along with land development intensity and road density. All three together matter for city compactness. As studied by Yao et al. (2022), large compact cities with high population density provide high urban economic efficiency.

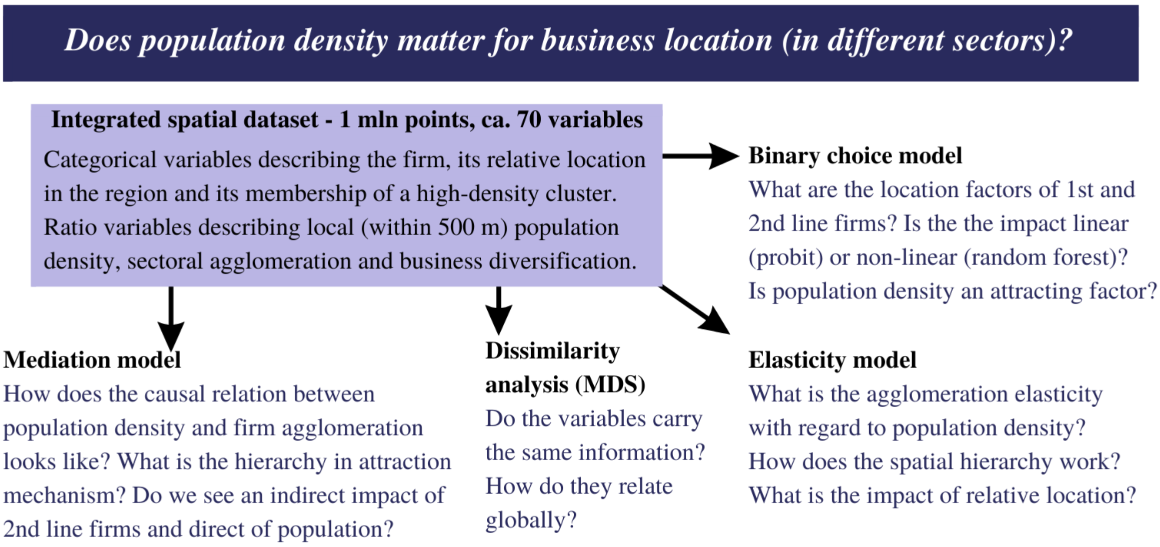
Immediate interaction between inputs and outputs follows Jacob's (1961) logic that local retail and services support not only other businesses but also are critical to the vitality of community life (Meltzer, 2016).

2. Methodology of modelling the impact of density and agglomeration externalities on the business location

Recent literature developed a few approaches to the model business location (overview in Carpenter et al., 2021), also conditioned by agglomeration externalities. However, none of these addressed issues raised in this paper: hierarchy of agglomeration externalities and non-linearity of relations. Thus, even though there are many existing solutions, they are not exhaustive and still need to be complemented.

We propose three general models: mediation model, binary choice model and elasticity model (Fig.1). Mediation model aims to detect the hierarchy of impact of population density and business agglomeration on business location decisions. The binary choice model is to detect the determinants of the location of 1st and 2nd line firms. The elasticity model is to assess the elasticity of business agglomeration related to the density of population and localised effects of a relative neighbourhood. All models test the hypothesis that there exist two channels of business attraction: through business agglomeration externalities (matching, sharing, learning mechanisms) and through population density externalities (attracting mechanism). The mediation model by design tests the existence of causal hierarchical relation of attraction mechanism without including control variables (*section 2.1*). Once the causal relation is proven to exist, binary choice models (probit and random forest) use controls to estimate the impact of density and agglomeration externalities on business location for 1st and 2nd line firms separately (*section 2.2*). The elasticity model is to capture the relation between business agglomeration and population density, controlling for relative location (*section 2.3*). These three models explain the existence of 1st and 2nd line firms in a given location – they use factors discussed in *section 2.5*. Estimated equations are discussed in *section 2.6*. The inclusion of many controls, besides being theoretically justified, requires statistical analysis. Dissimilarity analysis (*section 2.4*) replaces correlation analysis in checking if control variables bring different information, which is to avoid multicollinearity.

Figure 1: Flow chart of study design



Source: Own work

An important issue is to distinguish between agglomeration externalities and agglomeration phenomenon. Our approach is based on the agglomeration phenomenon (measured as a number of firms), while benefits – agglomeration externalities – stay implicit. This paper does not concentrate on finding what kind of benefits called agglomeration externalities appear for a given firm. However, it takes an implicit and comprehensive effect expressed by location – if firms behave due to the theory where some neighbourhood is profitable, and we observe they locate there, we assume those benefits appear. The same refers to externalities of population density – if a location next to people is profitable and firms locate there, one takes this implicit effect as existing.

The latest empirical literature on micro-geography of firms and agglomeration externalities (e.g. Dubé et al., 2016; Andersson et al., 2019) developed studies concentrating on local neighbourhoods – surroundings derived individually for each firm and distance-decaying interactions within a narrow radius of several hundred meters. This stream of research is to be supplemented with micro-data on the human population close to firms – to operationalise and understand the population density externalities.

2.1 Mediation model to test causality

The problem raised in this paper is the hierarchy of the impact of density and agglomeration externalities on business location. We hypothesise that population density determines the appearance of retail and service less-productive non-innovative 2nd line firms, which in turn attract the most productive and innovative 1st line firms. However, population density itself is also an attractor of 1st line firms. This logic of causality can be tested within the framework of the mediation effect (Barron & Kenny, 1986, Imai et al., 2010, Antonakis et al., 2010). Using the mediation model we establish a causal connection between population, agglomeration of 2nd line firms and 1st line firms (Fig.2). The core of the mediation model is a variable called the mediator, which channels the impact of the explanatory variable on the outcome variable. The total impact of the explanatory variable is a total of direct effect (ADE, *Average Direct Effect*) and indirect effect (ACME, *Average Causal Mediation Effect*). This indirect effect is a product of coefficients from two regressions: the first one, where the mediator is explained by the explanatory variable (eq.1) and the second one, where the outcome variable is explained by the mediator (eq.2):

$$mediator_i = \alpha_0 + a \cdot x_i + \varepsilon_i \tag{1}$$

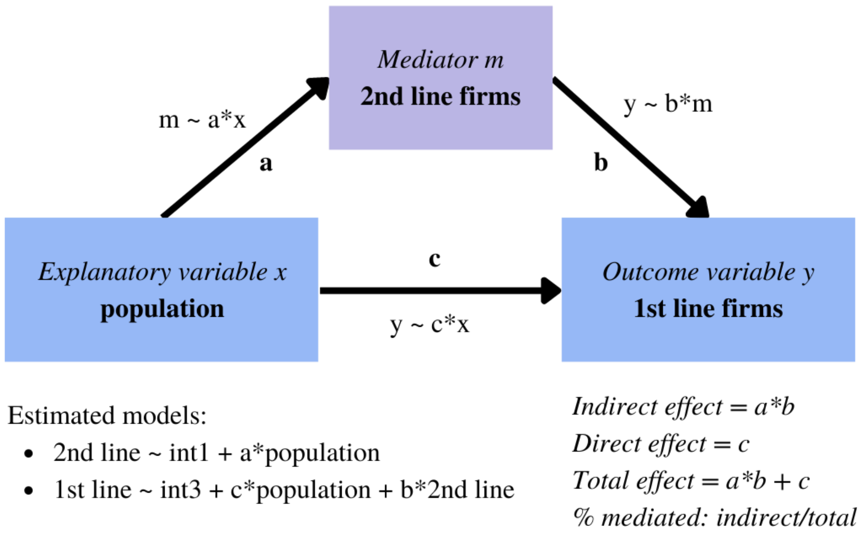
$$y_i = \alpha_1 + c \cdot x_i + b \cdot mediator_i + \varepsilon_i \tag{2}$$

which measure the direct effect (coefficient c), indirect effect (coefficients $a \cdot b$) and total effect ($a \cdot b + c$). The share of mediated effect is the ratio of indirect and total effects. In this case, the variables are agglomerations in a 500 m radius: $mediator_i$ is the number of 2nd line firms, y_i is the number of 1st line firms, and x_i is the number of inhabitants.

The motivation for this setting is as follows. High population density creates a large consumer base which becomes a beneficial location for the 2nd line firms (Fig.2 relation a), what is in line with the liveable city concept. In consequence, entrepreneurial potential stimulated by the increased number of 2nd line companies increases the probability of 1st line firms being opened (Fig.2 relation b). This relation goes one way only, as much higher capabilities and know-how are needed for opening a 1st line firm. Also, the convenience factor makes highly-skilled and well-earning workers locate next to everyday life facilities. Simultaneously, densely populated areas provide a higher talent pool that can serve as an employee base for 1st line companies. In turn, 1st line firms are more likely to co-locate near highly populated places to maximise the matching mechanism between creative individuals and innovative companies (Fig.2 relation c).

Some may argue however that this representation of the mediation effect is too simplistic and ignores the complex dimension of bi-directional feedback loops between these phenomena: that people attract business, and at the same time, the business attracts people? Potential endogeneity could arise if people settle down due to jobs in the 1st line firms or increased liveability offered by 2nd line firms. However, the emergence of this relation is highly restricted. The population density in the short term is limited by the available accommodations, preventing any potential impact of businesses on population density. Even in the long term, when the housing market could possibly adapt to the increased demand, it is still bounded by the land available for housing investments, which is a highly scarce resource in the urban space. Therefore the only impact that business can have on the population is on its quality rather than quantity – a process widely known as gentrification (Meltzer, 2016). The issue of endogeneity in the mediation model was tested with instrumental variables (IV) (Imai et al., 2010) (Appendix 6). We found that number of flats from the census 25 years earlier is a strong instrument (correlated with population and uncorrelated with business as it dates back to the communistic era), and the accompanying IV regression test evidence no endogeneity in this setting.

Figure 2: Mediation model



Source: Own work based on Baron and Kenny (1986)

2.2 Binary choice models: probit and random forest

Our approach assumes modelling of the decision process of business location. The entrepreneur decides on establishing a specific firm (1st line, 2nd line, in a given sector) first and, secondly, looks for the location. The location decisions are binary (yes/no) for a given place. The model is to assess how much a given location is attractive for specific firms. Operationally, the binary dependent variable (*If_to_locate_in_xy*) expresses if the firm in a given location (already known) belongs to a given type (also already known). One estimates a separate model for 1st line firms and individual models for 2nd line firms by sectors (agriculture, production, construction, service). Models for 2nd line firms do not consider the 1st line firms. This approach does not flatten the impact hierarchy and allows for wide control of the local neighbourhood.

The binary choice was modelled with a classical econometric estimation (probit) and supervised machine learning (random forest) (e.g. Kopczewska, 2022). Both approaches used the same formal equation:

$$dummy_{locating \in xy} = f(setofdummyvariables, setofstandarisedvariables) + \varepsilon \qquad (3)$$

Methodological details of both models are in Appendix 4.

2.3 Elasticity model

The elasticity model is to detect the relationship between population density and business agglomeration in a small spatial scale – a radius of 500 m. Additionally, it can identify local spatial deviations from the main trend resulting from a specific location. Elasticity models have been well-developed in the last two decades, however, they are mostly focused on productivity or wages and rarely considered spatial heterogeneity. The general form of the model is as follows:

$$log(agglomeration) = f(log(density), setoflocationdummies) \qquad (4)$$

where *log(agglomeration)* is the logarithm of a number of firms located in a radius of 500 m from the analysed firm, *log(density)* is the logarithm of inhabitants in a radius of 500 m, while a set of dummy variables for location explains if a point is in the radius of 10, 25 or 50 km from the centre of cities of different size or if it belongs to a high-density cluster of firms or population. The model is estimated with OLS as the relation *log(y)~log(x)* turned out to be linear. Beta coefficient at density ($\beta_{density}$) is interpreted as elasticity of agglomeration because of density. In general, one should expect an elasticity coefficient <1, to assure that agglomeration does not grow to infinity, especially in densely populated areas. However, an elasticity coefficient >1 may appear in non-overcrowded areas highly attractive for business, e.g. suburbs of bigger cities. Dummy variables reflecting relative location can detect spatially localised variations in agglomeration elasticity.

2.4 Dissimilarity analysis

Instead of a typical correlation analysis, we have conducted Multidimensional Scaling (MDS). There are a few reasons for that: one is that the dataset contains many dummy variables (for which Pearson correlation does not work) and the other is that correlation analysis refers to selected pairs only, neglecting the joint analysis with other variables in the dataset. MDS is an unsupervised machine learning method to reduce dimensions – it can project the dataset with many more dimensions (variables) into two dimensions (*x,y*) and plot it. MDS calculates the multidimensional distance between objects (variables) in the original *k*-dimensional space. Secondly, it assumes *k* points in the new 2D projection and calculates distances between those points (more in Appendix 4).

Outcome – a two-dimensional bubble plot – illustrates the relative similarity of variables (*xy* location), and the impact of a selected variable on the whole setting (bubble size). The bigger the bubble the higher the *Stress Per Point* (*spp*) which reflects the difference between the given variable compared to the other variables considered jointly. Variables with low *spp* are similar to each other and fit the model well. Variables with high *spp* carry very different information in relation to all other variables, even if they reduce the overall fit of the model.

2.5 Explanatory factors

Explanatory variables follow the theory of location decisions. There are at least seven groups of variables which should be included in the model specification. These are as follows. Operational details are in Tab.1.

1. **Relative peripherality** – location decisions are driven by relative distance to the cities within the region or distance to the city centre within the city. In many research, the most attractive locations are in Central Business District (e.g. Pan et al., 2018). Typically, models include distance to the centre of a given city. However, in the case of a region with a dense settlement network and many urbanised areas, this approach is not efficient. First, firms located between core and satellite cities benefit from both urban areas, thus classifying them as gravitating towards one or another city might be misleading. Secondly, non-core cities are often homogeneous and for many firms, it does not matter if they locate close to city A or B if their size is similar. Therefore, we use a set of dummies to express if a given firm is within a specific range (10 km, 25 km, 50 km) from cities of different sizes (core 1 mln+, midsize 100K+, regional 50K+, local big 30K+, local small 15-30K). The model includes therefore 15 variables (e.g. *dist_core_10*) – for 3 different distance radii to 5 types of cities.

   That allows for the modelling of hierarchical relative location and distinguishing between core urban locations, suburban locations around the main city, locations in second-tier cities and locations outside of main cities. The expectation is that some business sectors which run land-based activities (such as agriculture, forestry, and sectors with large storage areas) prefer peripheral locations. On the other hand, knowledge-intensive sectors concentrate in big cities, due to matching, sharing, learning and attracting mechanisms. There are also public services, that must be provided throughout the entire inhabited territory. This suggests that relative location may be significant/important in a diverse way.

2. **Density externalities** – population density is a core of this study and, until now has rarely been used in explaining business location. The assumption is that the higher the population density in the surrounding, the more attracting power for new businesses which use human capital or are based on sales and/or human interactions. This model studies population density in a narrow radius of 500 meters from the firm location (*locPdens*). Recent progress in world-wide population data availability on low aggregation levels (like 1 km^2 grid) allows for high-precision analysis. Some selected studies used population density but mostly aggregated at the municipality level (Backman & Karlsson, 2017 for Sweden, Guimarães et al., 2000 for Portugal, Elgar et al., 2009 for Canada, Balbontin & Hensher, 2019 for Australia; Carlino et al., 2007 for patenting in USA) – even though, they evidenced its significant and positive impact.

3. **Agglomeration externalities** – business agglomeration is a fundamental variable in analysing firms' locations. It measures the number of firms in the surrounding, assuming that the more firms located in the neighbourhood, the higher the attractiveness of the location. For consistency, it should be measured in the same radius as population density (500 meters). This approach offers the possibility of deep decomposition into sectors (*locAggAgri, locAggProd, locAggConstr, locAggServ*) and within different radii (for attenuation of externalities what is not analysed in this paper) (see e.g. Dubé et al., 2016).

4. **Urbanisation externalities** – the literature is not clear on the operationalisation of the concept of urbanisation externalities. There are some studies (e.g. Klein & Crafts,

2020) that equalise this term with Jacobian agglomeration and take it as the number of firms in the surroundings from the other sectors than the analysed one. However, we see such a definition as closer to local specialisation and diversity, covered in separate points below. In this study, we claim that urbanisation externalities are the effects coming from the high density of population and/or business. This is about the critical mass of people and/or firms required to trigger urbanisation effects. That is why it differs from simple population density or business density and is instead the binary information if a given location belongs to the territory with the highest density or not, referred to a wider surrounding. We treat it as a relative measure by using the DBSCAN clustering algorithm, which indicates locations with the highest density within the analysed territory (Appendix 2). This approach is suitable for point data and simultaneously very flexible and precise, as each point location is classified as a high-density core or as low-density noise. This study considers two variables based on DBSCAN classification: one detecting if an observation belongs to a population high-density cluster (*COREpopul*), and the second for the observation belonging to a business high-density cluster (*COREfirms*).

5. **Local industrial specialisation** – a local neighbourhood of a firm should be assessed not only as mass but also through the structure. The composition of business surroundings has appeared in the literature for a long time (Kopczewska et al., 2017), and Location Quotient (LQ) is often used as a measure of local industrial specialisation (Pominova et al., 2022, Artz et al., 2020). It relates two proportions of business in a given sector compared with all establishments, one in the narrow surrounding and the second over the whole territory. Formally, it is expressed as:

$$LQ_i = \frac{\frac{\sum N_{S,radius}}{\sum N_{radius}}}{\frac{\sum N_{S,region}}{\sum N_{region}}} \tag{5}$$

where subscript S stands for the sector, *radius* for the local neighbourhood, and *region* for the whole territory, and N is the number of firms. LQ can be understood as the ratio of self-sector shares – in the analysed radius and in the whole dataset. Values of $LQ≈1$ suggest that local sectoral composition is similar to general, and no local specialisation is observed. Values of $LQ>1$ (practically $LQ>1.15$) indicate a concentration of similar businesses in the surrounding, while opposite values of $LQ<1$ (practically $LQ<0.85$) suggest sectoral loneliness of the firm. In this model, we include LQs for the sector of the analysed firm in a 500 m radius (*locLQ*).

6. **Local industrial diversity** – the local composition of a business neighbourhood can be approached with the Herfindahl-Hirsch index (HH) within a specified radius from the given firm (e.g. Artz et al., 2020). HH index measures the degree of industrial monopolisation, or oppositely, competition. HH index is expressed as:

$$HH = \sum_{radius} s_k^2 \tag{6}$$

where s_k is the share of the firm (by sales, employment etc.) in a given sector across the whole territory. This measure calculates the sum of the overall shares for local neighbourhoods for a given sector (as observation). Interpretation of HH values is in commonly accepted thresholds: $0<HH<0.15$ for high competition and no concentration (high diversity), $0.15<HH<0.25$ means moderate concentration as there exist from 4 up to 7 firms on the market with equal shares (moderate diversity), while values $0.25<HH<1$ suggest high concentration or even monopoly (low diversity). In this model, we include HHs for the sector of the analysed firm (*locHH*).

7. **Local competition** – business location decisions may be driven by a willingness to locate next to important business players. These are usually large and well-established companies, employing hundreds of workers and possibly maintaining a high market share. As evidenced by Porter (2008), co-location with big firms may be profitable. This concept can be measured with the agglomeration of big firms in a local neighbourhood of each firm, here in a radius of 500 m (*locBIG).*

2.6 General forms of equations for models

Each of three models requires a separate specification to capture the effects of interest.

Mediation model, a form of a structural equation (eq.1-2), aims to detect a hierarchy in density and agglomeration:

$$locAgg2nd_i = \alpha_0 + a \cdot locPdens_i + \varepsilon_i \quad (7)$$

$$locAgg1st_i = \alpha_1 + c \cdot locPdens_i + b \cdot locAgg2nd_i + \varepsilon_i \quad (8)$$

where for each observation (firm) the following are measured within a 500 m radius: the number of 1st line firms (*locAgg1st*), the number of 2nd line firms (*locAgg2nd*) and a number of inhabitants (*locPdens*).

Probit and random forest models (eq.3)**,** applied to explain location decisions, use a dummy as a dependent variable and factors described above as explanatory variables:

If_to_locate_in_xy ~	*Decision if to locate in given place*	
dist (core, midsize, regional, localbig, localsmall) +	*(1) relative peripherality*	
+ locPdens +	*(2) density externalities*	
+ loc_agglom (agr, prod, constr, serv) +	*(3) agglomeration externalities*	
+ COREpopul + COREfirms +	*(4) urbanisation externalities*	(9)
+ locLQ +	*(5) local industrial specialisation*	
+ locHH +	*(6) local industrial diversity*	
+ locBIG	*(7) local competition*	

The detailed form of the equation is given as:

$$\begin{aligned} dummy_{firm} = {} & \beta_0 + \beta_1 \cdot dist_{core10,i} + \beta_2 \cdot dist_{midsize10,i} + \beta_3 \cdot dist_{regional10,i} + \beta_4 \\ & \cdot dist_{localbig10,i} + \beta_5 \cdot dist_{localsmall10,i} + \beta_6 \cdot dist_{core25,i} + \beta_7 \\ & \cdot dist_{midsize25,i} + \beta_8 \cdot dist_{regional25,i} + \beta_9 \cdot dist_{localbig25,i} + \beta_{10} \\ & \cdot dist_{localsmall25,i} + \beta_{11} \cdot dist_{core50,i} + \beta_{12} \cdot dist_{midsize50,i} + \beta_{13} \\ & \cdot dist_{regional50,i} + \beta_{14} \cdot dist_{localbig50,i} + \beta_{15} \cdot dist_{localsmall50,i} + \beta_{16} \\ & \cdot locPdens_s + \beta_{17} \cdot locAgg_{agri.s} + \beta_{18} \cdot locAgg_{\prod,s} + \beta_{19} \cdot locAgg_{constr,s} \\ & + \beta_{20} \cdot locAgg_{serv,s} + \beta_{21} \cdot CORE_{firms} + \beta_{22} \cdot CORE_{popul} + \beta_{23} \cdot locLQ_s \\ & + \beta_{24} \cdot locHH_s + \beta_{25} \cdot locBIG_s + \varepsilon_i \end{aligned} \quad (10)$$

where $dummy_{firm}$ is a binary variable indicating if a given firm belongs to a selected group (y=1) or not (y=0) – groups defined as 1st line high-tech firms, 2nd line agriculture, 2nd line production, 2nd line construction and 2nd line service. Variables were defined as in section 2.5, ending .s is for standardised variables (other variables are dummies), the dataset for 2nd line firms excludes 1st line firms.

Elasticity model tests the relation between logarithms of business agglomeration (dependent variable, *log(locAggTotal)*) and population density (explanatory variable, *log(locPdens)*), using additional controls – relative location dummy variables defined in the "relative peripherality" group (eq.4). The estimated equation is in the form:

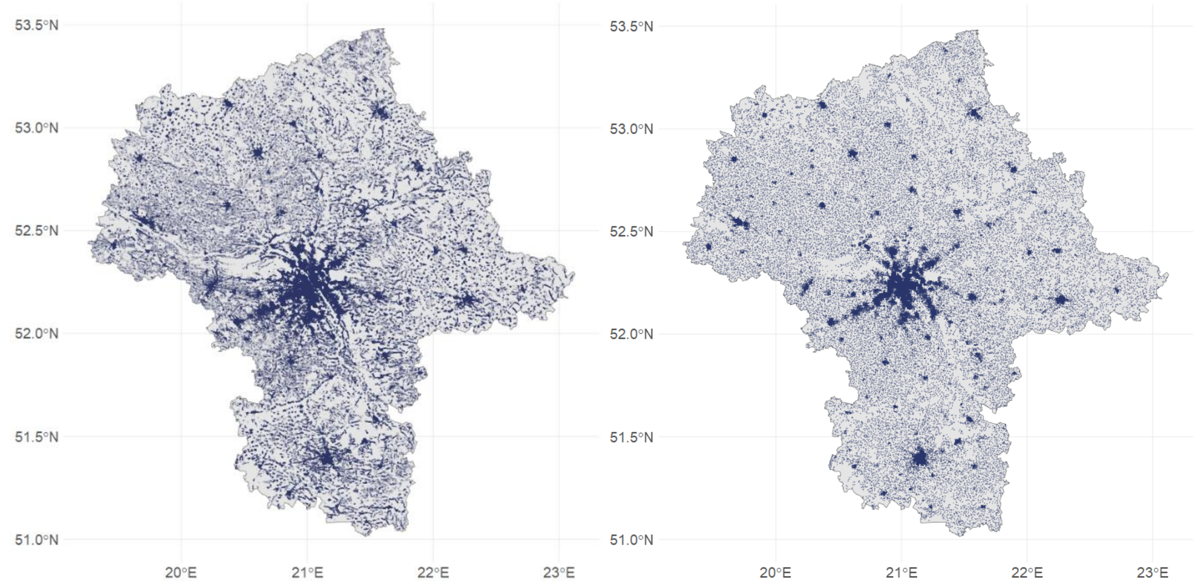
$$
\begin{aligned}
log(locAggTotal) = {} & \gamma_0 + \gamma_1 \cdot log(locPdens) + \\
& + \beta_1 \cdot dist_{core10,i} + \beta_2 \cdot dist_{midsize10,i} + \beta_3 \cdot dist_{reg10,i} + \beta_4 \cdot dist_{locbig10,i} + \beta_5 \\
& \cdot dist_{locsmall10,i} + \beta_6 \cdot dist_{core25,i} + \beta_7 \cdot dist_{midsize25,i} + \beta_8 \cdot dist_{reg25,i} \\
& + \beta_9 \cdot dist_{locbig25,i} + \beta_{10} \cdot dist_{locsmall25,i} + \beta_{11} \cdot dist_{core50,i} + \beta_{12} \\
& \cdot dist_{midsize50,i} + \beta_{13} \cdot dist_{reg50,i} + \beta_{14} \cdot dist_{locbig50,i} + \beta_{15} \\
& \cdot dist_{locsmall50,i} + \varepsilon_i
\end{aligned}
\tag{11}
$$

where *log(locPdens)* identifies the elasticity of agglomeration with respect to population, while dummy variables for distance from cities are to detect local instability of elasticity.

3. Empirical verification of density-based location decisions model

Dataset: This research uses full sample (statistical population) data for businesses and inhabitants. Business data contain almost 1 mln geo-localised firms for the Warsaw region in Poland (formally Mazovian NUTS1 voivodship) (Fig.3), what covers all firms located in this region. Data for 2012 were retrieved from the official register of polish business entities (REGON) and included address, main sector (indicated with A:U letter and 5-digit code) and employment (in groups 1:5) (Appendix 1). Data were geo-coded from postal addresses to latitude /longitude. Population data were obtained from the official census published in 2012 in a 1 km x 1 km grid with information on the number of inhabitants, age and gender structure in each grid cell (Statistics Poland, 2012). The Mazovian region has a population of ca. 5.4 million people and a territory of 35'558 km^2. It gives an average population density of 150 persons/km^2 (varying from 0 to 21'500 persons/km^2) and a business density of 28 firms/km^2 (varying from 0 to 16'800 firms/km^2).

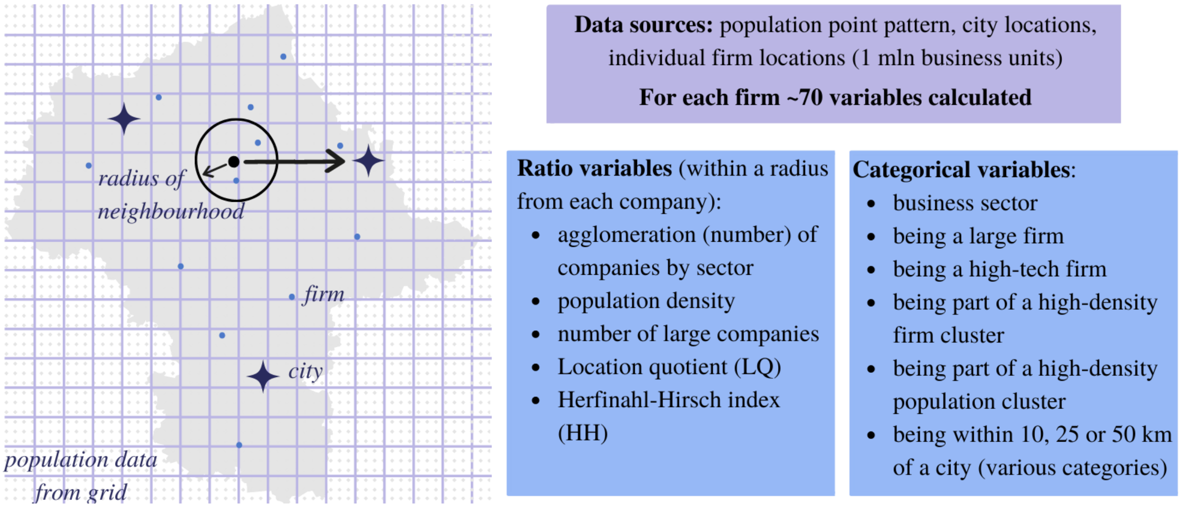
Figure 3: Spatial data used in analysis: a) business location, b) population location



Source: Own work

Spatial granulation of the data matters for the precision of the estimation and how the variables in the model are constructed (Fig.4). The literature primarily uses two types of data: individual data attached to geographical coordinates, and aggregated data attached to the region. Individual data give the highest flexibility as they can be aggregated and summarised in all possible ways. This kind of data was used by e.g. Dubé et al. (2016) in studies for Canada. Aggregated data are used in the case when individual data are not available (e.g. Andersson et al. 2016 in cells 1km^2 for Sweden). For individual data, the most convenient aggregation method is a circular territory with a radius of a few hundred meters (e.g. Dubé et al., 2016) (Appendix 3). In this study, we computed the explanatory variables (denoted as *loc*) in the circular neighbourhood of 500 meters radius. This radius relates to the observed points density and allows for a non-zero neighbourhood for a vast majority of points and variables. Population data available as a grid were transformed into a point pattern by sampling randomly an equivalent number of points (here 1 point for every 100 persons) within each 1 km^2 grid box. The location error at 1km x 1km resolution ranges from a few up to a few hundred meters (ca. 10 minutes' walk), which remains insignificant for statistical analysis of larger areas (Appendix 3).

Figure 4: Spatial dimensions of data



Source: Own work

Based on those two sources of data and the location of cities within the analysed region (Appendix 2) we derived variables (Tab.1) as described in Section 2. The statistical overview of data (Appendix 3) provides some general observations. Almost all firms are located max. 50 km from a small city, and every third firm not further than 10 km from the core city (33%). Firms prefer locations next to regional cities (50-100K) than mid-size cities (100K+). 1st line firms and big firms have denser surrounding than 2nd line firms and mostly are located in high-density clusters of population. 1st line firms in 57% are located in a core city (r=10 km) and 27% in its fringe (r=25 km), while the remaining 16% are in other locations without a clear pattern. 1st line firms differ from 2nd line firms (by sectors) in a majority of variables.

Table 1: Variables used in the analysis

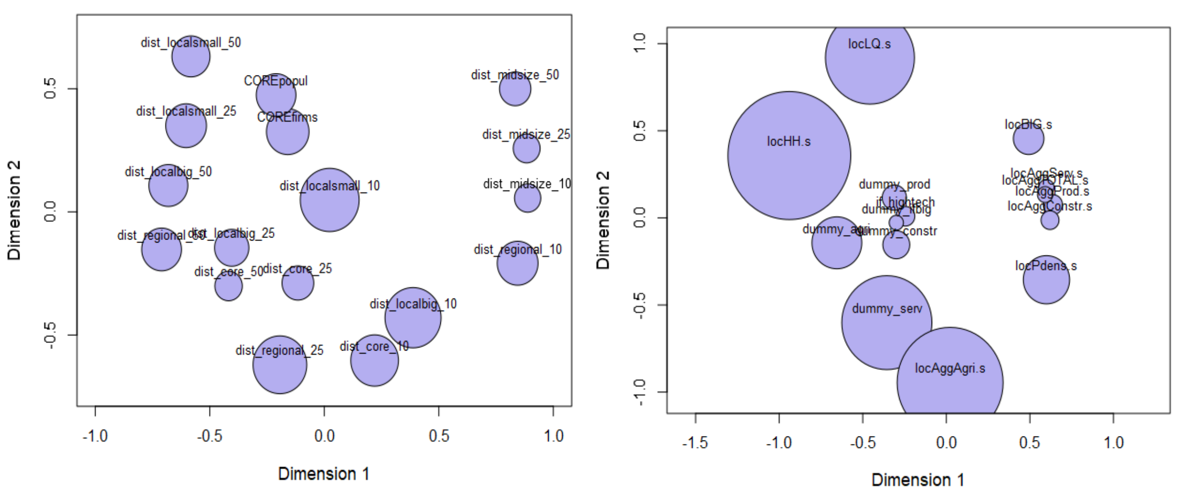
Type	*Variable name*	Description
location	*dist_core_10, dist_core_25, dist_core_50, dist_midsize_10, dist_midsize_25, dist_midsize_50, dist_regional_10, dist_regional_25, dist_regional_50, dist_localbig_10, dist_localbig_25, dist_localbig_50, dist_localsmall_10, dist_localsmall_25, dist_localsmall_50*	Dummy variable for each firm: 1 if a firm is located in a radius (10, 25 and 50 km) from the city centre. We considered 39 cities in 5 population size categories that are located within NUTS1 Mazovian region: core 1mln+ (1 city, Warsaw), midsize 100K+ (2 cities), regional 50K+ (4 cities), local big 30K+ (9 cities), local small 15-30K (23 cities).
	COREfirms, COREpopul	The dummy variable defines if a given point belongs to a high-density cluster from DBSCAN (of a radius of 0.03° what is equivalent to ca.3,3 km and minPts=75 firms / 500 persons).
neighbourhood	*locAggA, locAggB, .., locAggU*	A number of firms from a given sector (A:U, Appendix 1) calculated in a radius of 500 m from a given firm
	locAggAgri, locAggProd, locAggConstr, locAggServ,	Aggregated local sectoral agglomeration values: Agriculture: sector A; Production: sectors B, C, D, E; Construction: sector F; Service: sectors G, H, I, J, K, L, M, N, O, P, Q, R, S, T, U
	locAggTotal	A total number of firms calculated in the direct neighbourhood of 500 m of a given firm
	locHH	Herfindahl-Hirsh index calculated in 500 m radius
	locLQ	Location Quotient calculated in 500 m radius
	locBIG	The number of firms classified as big in the direct neighbourhood of 500 m of a given firm

	locPdens	A number of inhabitants in the direct neighbourhood of 500 m of the given firm
features	*Address, latitude and longitude*	Postal address of the firm from the REGON register, recoded into geo-coordinates
	SEC_PKD7, PKD7	Sectoral classification of firm's main activity from the REGON register
	empl, gr_empl	Employment size. *Gr_empl* reports five employment classes: up to 9 persons (gr.1), 10-49 persons (gr.2), 50-249 persons (gr.3), 250-1000 persons (gr.4) and 1000+ persons (gr.5). *Empl* gives the approximate mid-value of groups, respectively 5,30, 150, 600 and 1500
	dummy_if_highetch	Dummy variable if a firm can be classified as high-tech business
	dummy_if_big	Dummy variable if a firm can be classified as big (employment above 250 persons, groups 4 and 5)
	dummy_agri, dummy_prod, dummy_constr, dummy_serv	Dummy variable if a firm belongs to one of four main sectors (see details for locAggAgri, locAggProd, locAggConstr, locAggServ)

For estimation, non-dummy variables were standardised. More details in Appendix 1 and 2
Source: Own work

MDS (*multidimensional scaling*) dissimilarity analysis reveals that among location-related dummy variables (belonging to a high-density area and being within the radius of 10 km, 25 km and 50 km from big, midsize, regional and small cities) each variable carries different information (Fig.5a) – the bubbles are well separated. Being located within a core city (*dist_core_10*) and being located in the area of a high-density of firms or people (*COREfirms, COREpopul*) is not the same information and all of them are very influential (due to the large size of the bubble). In the group of neighbourhood and self-characteristics variables (Fig.5b) it is visible that population density in the surrounding of firms (*locPdens*) matters a lot and does not overlap with a density of sectoral agglomeration (*locAggAgri, locAggProd, locAggConstr, locAggServ*) or surrounding of big firms (*locBIG*). MDS analysis clearly shows that the inclusion of population density is an important factor in understanding business location and that all the variables taken into account bring new diverse information.

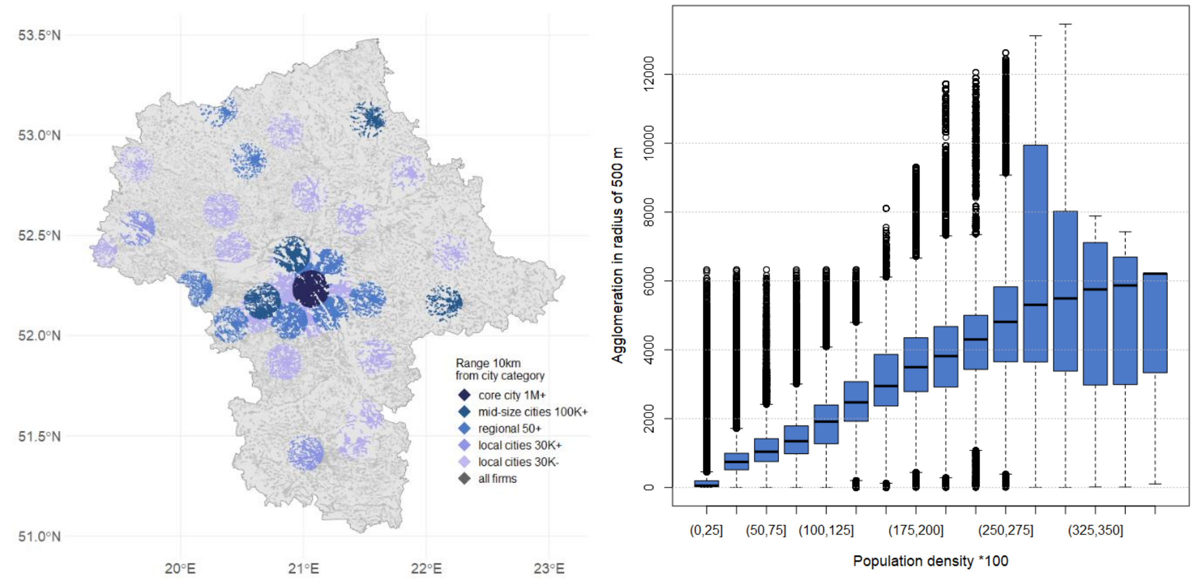
Figure 5: Dissimilarity of variables: a) for distance, b) for neighbourhood



Note: Dimensions 1 and 2 are MDS projections and illustrate proximity of variables (in pairs). The circle's size (bubble) is the stress-per-point (spp) – the greater the more diverse information included in the variable.
Source: Own work

Statistical inspection shows that around 70% of firms are located within 10 km from the city centre (Fig.6a), while the remaining 30% is much more peripheral. There exist a clear relationship between population density and business agglomeration (Fig.6), which is non-linear and non-stationary (Fig.6b). Under-trend non-linearity appears at the highest density – this can be justified by the limited capacity of available real estate in locations massively inhabited by people and firms. Non-stationarity (increased variance) is also typical for high-density areas – the diversity of highly populated areas in terms of attracted business is much higher than in low-density locations. This suggests the existence of additional factors (e.g. feedback loops, knowledge spillovers, diseconomies of urbanisation, etc.) in high-density areas in attracting business to that place. It also suggests the existence of diverse spatial structures.

Figure 6: Relative spatial distribution of firms: a) to city centres, b) to population density



Source: Own work

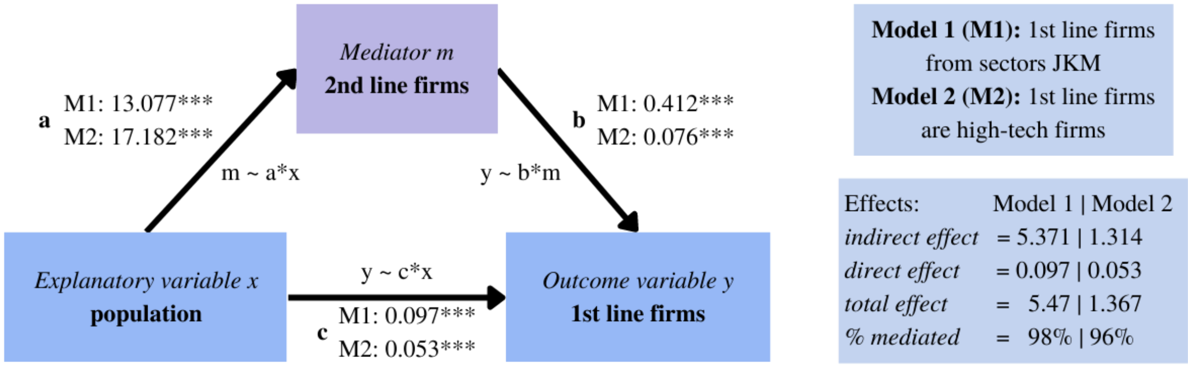
The mediation model aimed to test if there exists mediation (indirect) and direct effects between business location and inhabiting population. We verified this hypothesis threefold: with two models on point data (Tab.2, Fig.7), and using Poland-wide municipalities data (Appendix 6). All approaches show a significant mediation effect as assumed, justifying the design of further analyses. Moreover, the general model for municipalities was extended to include instrumental variables to eliminate a potential reverse direction reaction (Appendix 6) – the results ensured the validity of the causal relation between population, 1st and 2nd line firms while proving that there is no endogeneity issue in the mediated relation.

Mediation models on point data (eq.7-8, Tab.2) use firm as the unit of analysis and its neighbourhood within a radius of 500 m. The explanatory variable in both models is the population density in firms' surroundings. In the first model, the mediator variable is the local agglomeration of 2nd line firms (firms from all sectors except J, K and M – details in Appendix 1), while the outcome variable is the local agglomeration of 1st line firms (from J, K and M sectors). In the second model, the mediator variable is the local agglomeration of non-high-tech firms, while the outcome variable is the local agglomeration of high-tech firms (Tab.1). In both cases mediation effect and direct effect are significant. The size of the mediation effect was estimated at ca. 96-98%. Results from the mediation model show that the population density is a direct cause of the firm agglomeration, while the effect on the 1st line firm location is mediated by the 2nd line companies creation.

Table 2: Mediation effects

Mechanism	Coefficients from model 1 1st line firms from J,K,M sectors	Coefficients from model 2 1st line firms are high-tech firms
Indirect effect (ACME, *Average Causal Mediation Effect*)	5.371*** (13.077*** · 0.412***)	1.314*** (17.182*** · 0.076***)
Direct effect (ADE, *Average Direct Effect)*	0.097***	0.053***
Total effect	5.468***	1.367***
% mediated	98%	96%

Figure 7: Mediation effects



Source: own work

Probit model and random forest model were estimated using eq.9-10 (Tab.3, Appendix 4-5). Non-dummy standardised data allow comparison of coefficients between models and interpretation of changes by 1 on a standard deviation scale.

The general conclusion from the **probit model** estimation can be summarised as follows. **Population density** (*locPdens*) matters for the business location of all firms. Its impact is positive in the case of 1st line high-tech firms and 2nd line service sector due to sectoral preferences towards highly urbanised areas. The firms from other sectors (agriculture, construction, production) locate in less dense areas (negative impact, preferred density is below average). Other density-related variables are also relevant, including dummies from DBSCAN if the location belongs to a high-density core or low-density noise. It applies both to the density of inhabitants (*COREpopul*) and firms (*COREfirms*). 1st line firms and 2nd line production, construction and service prefer high-density locations. Interestingly, population density is the crucial factor here – for 2nd line production firms it matters way more than being in a high-density cluster of firms. In this example, one can see the importance of the attraction mechanism of the population across different business sectors. **Sectoral agglomeration** is significant in the Marshallian way (firms surrounded by the same sector) – the strongest effect is in 2nd line service (l*ocAggServ*). Jacobian relations understood as a diversified neighbourhood, are also visible – nearby agglomeration from other sectors is significant, but mostly with negative (*loccAggAgri, loccAggProd, loccAggConstr, locAggServ*).

Interestingly, 1st line firms and agriculture tend to create high local specialisation (*locLQ*), while other firms prefer the local LQ below the average LQ (more diversified environments), as shown by negative coefficients. This suggests strongly Marshallian agglomeration in 1st line firms and agriculture and more Jacobian composition in other sectors. High industrial diversity (*locHH*) appears in 2nd line agriculture and construction due to many big firms. Coefficients are approaching zero for 1st line firms, and 2nd line production and service, which suggests many small firms on the market. There is no clear pattern corresponding to the proximity to big firms. Negative coefficients (*locBIG*) for 1st line and 2nd line service models should be associated with the highest average values in these sectors (Tab.3) and a downward correction towards the general average. 1st line firms, 2nd line service and production firms tend to locate around cities of any size. For service, there is a clear preference to locate in proximity to bigger settlements (within the radius of 10, 25 or 50 km around core, midsize or regional cities), while the local cities (below 45K inhabitants) are much less preferred. The production sector prefers midsize cities (100K+) within any radius and or local cities as second-best. The agriculture sector in general escapes from any city, tending towards peripheral and rural areas. Coefficients for the construction sector are very weak, what may suggest no strict preferences. Estimation evidence that location attractiveness increases in the surrounding of any type of city, except for agriculture, which deliberately seeks out peripheral areas.

Results from the location choice model support the core concept of this paper, showing how population density matters for different business sectors. Although e.g. service and agriculture sectors have many different preferences towards the proximity of deeply populated areas, population density plays an undeniably important role in any business location.

Table 3: Estimation results of probit models for 1st and 2nd line firms

Variable	**1st line firms**	**2nd line Agriculture**	**2nd line Production**	**2nd line Construction**	**2nd line Service**
Intercept	-2.86***	-372.91***	1.91***	-1.71***	-0.79***
locPdens.s	0.02***	-0.43***	-0.09***	-0.05***	0.15***
locAggAgri.s	0	0.18***	-0.02***	-0.02***	0.02***
locAggProd.s	0.01	-0.02	0.34***	-0.05***	0.01
locAggConstr.s	-0.10***	0.59***	-0.07***	0.33***	-0.30***
locAggServ.s	0.11***	0.85***	-0.28***	-0.32***	0.32***
locBIG.s	-0.02***	1.20***	0	0.07***	-0.02***
locHH.s	-0.07***	-17615.15***	0.03***	-1.56***	-0.01***
locLQ.s	0.13***	0.68***	-0.14***	-0.33***	-0.20***
COREfirms	0.25***	-0.37***	0.01	0.02	0.59***
COREpopul	0.47***	-0.29***	0.21***	0.10***	0.38***
dist_core_10	0.06***	0.15***	-0.02**	-0.01	0.04***
dist_midsize_10	0.04	0.06***	0.08***	-0.01	0.12***
dist_regional_10	-0.01	-0.22***	0.02*	0.01	0.03***
dist_localbig_10	-0.11***	-0.06***	0.03***	0.02***	-0.04***
dist_localsmall_10	0.03***	-0.01	0.02**	0.03***	-0.04***
dist_core_25	0.19***	-0.16***	0.07***	0.01	0.23***
dist_midsize_25	0.12***	-0.15***	0.13***	0.11***	0.12***
dist_regional_25	0	-0.07***	0.03***	0	0.09***
dist_localbig_25	0.16***	-0.18***	0.06***	0.06***	0.17***
dist_localsmall_25	-0.01	-0.10***	0.01	0.05***	0.07***
dist_core_50	0.15***	-0.23***	0.10***	0.08***	0.11***
dist_midsize_50	0	-0.29***	0.06***	0.06***	0.13***

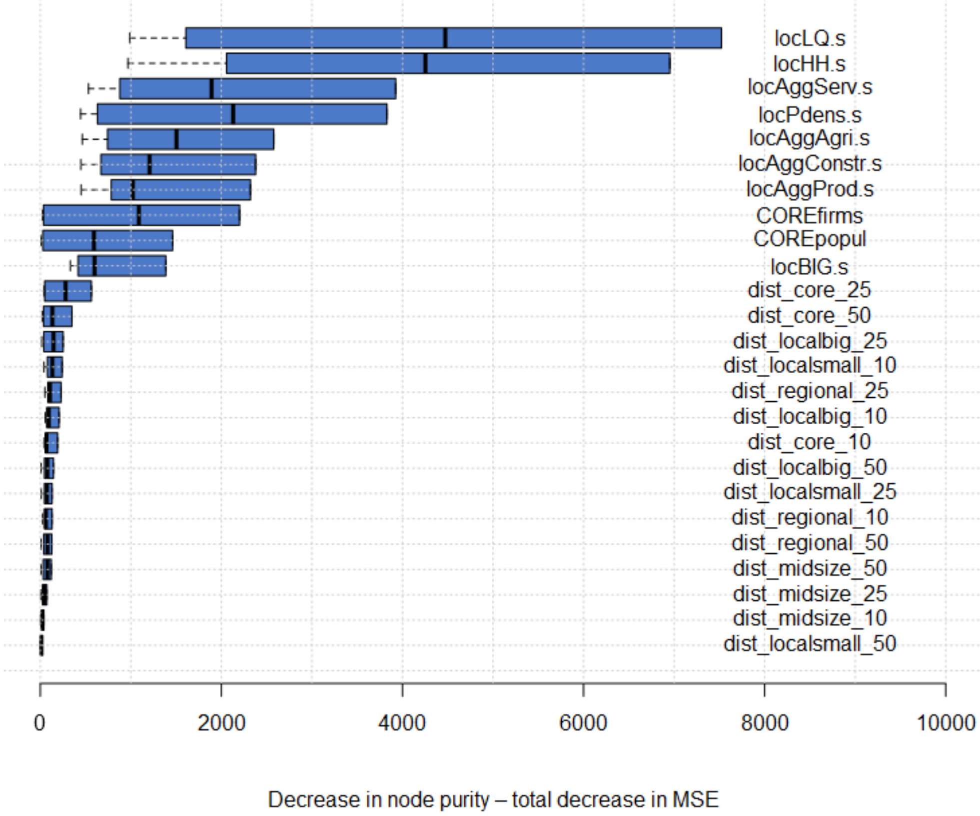
dist_regional_50	0.07***	0.10***	-0.06***	-0.09***	-0.04***
dist_localbig_50	-0.03	-0.05***	-0.02**	0.05***	0.07***
dist_localsmall_50	0.11*	-0.01	0	-0.07**	-0.09***
AIC	322492.2	424002.8	412808.3	480026.7	969777.9
McFadden R2	0.076651	0.613474	0.027648	0.052229	0.236849
BIC	322799	424308.5	413113.9	480332.3	970083.5
Log Likelihood	-161220	-211975	-206378	-239987	-484863
Deviance	322440.2	423950.8	412756.3	479974.7	969725.9
Num. obs.	983719	941378	941378	941378	941378

*P-value note: ***<0.001<**<0.01<*<0.05*

We verified the non-linearity present in spatial relations by estimating a **random forest** model for the same set of equations (eq.9-10). Supervised machine learning models are presented by the importance of variables (Fig.7, Appendix 4). Explanatory factors vary between the models, confirming that there is no unique set of location decision factors and that the intrinsic characteristics of business determine which set of location determinants matters. Population density – the core issue of this paper – is relevant in all models. Its importance is highly evident both in population density measured as the number of inhabitants around the firm (*locPdens*) as well as in a dummy if a firm is located in a high-density cluster of firms (*COREfirms*) and population (*COREpopul*). A comparison of probit and random forest models shows that they are highly consistent – indicating that the results are robust and insensitive to estimation methods.

The quality of predictions from the random forest models varies between them but remains high and reliable across all scenarios (Appendix 5). OOB error varies from 1.65% to 7.75%. The model predicted the location of service and agriculture firms very well (for 97%) and much worse in the production and construction sectors (for 45%). This result is consistent with the logit model results which showed strong patterns with respect to population density for agriculture and service sectors, while a bit more flexible relations were presented for the production and construction firms. Therefore, both probit and random forest models show that population density is a key factor for explaining the business location, irrespective of the differences across individual sectors.

Figure 7: Variable importance from Random Forest model for 1st line and 2nd line firms models



Source: own work

Elasticity model (eq.11) gives unambiguous results. The elasticity of agglomeration with respect to population density is around 0.76-0.79, which is the expected magnitude (Tab.4). This means that with a doubling population density, the agglomeration of firms increases by ca.76%. The coefficient below 1 is a natural consequence of congestion effects. Population and business cannot grow into infinity in highly congested areas. The reduction in available real estate that accompanies the agglomeration patterns extinguishes the multiplication effects in people–firms relations. The general pattern with regard to population is uniform – both 1st and 2nd line firms have similar values of elasticity, with a slightly stronger impact of population on the innovative companies. The elasticity of agglomeration is highly localised – the size of the city and distance to the centre matter. The model indicates a huge business potential in suburbs (up to 25 km) of almost all cities – coefficients for location dummies (*dist_..._25*) when summed up with the elasticity coefficient make the elasticity >1, which means stronger location potential for firms than people in those areas.

Table 4: Elasticity model with localised effects

Variable	Pooled data	1st line firms	2nd line firms
(Intercept)	3.27***	3.07***	3.27***
log_locPdens	0.76***	0.79***	0.76***
COREfirms	0.11***	0.01	0.12***
COREpopul	-0.12***	-0.03	-0.12***
dist_core_10	0.05***	0.10***	0.05***

dist_midsize_10	0.24***	-0.26***	0.24***
dist_regional_10	-0.43***	-0.18***	-0.44***
dist_localbig_10	-0.09***	-0.13***	-0.09***
dist_localsmall_10	0.13***	0.23***	0.12***
dist_core_25	0.74***	0.58***	0.74***
dist_midsize_25	0.40***	0.65***	0.40***
dist_regional_25	0.09***	-0.02	0.10***
dist_localbig_25	0.26***	0.57***	0.25***
dist_localsmall_25	0.19***	0.53***	0.18***
dist_core_50	0.10***	-0.04	0.10***
dist_midsize_50	-0.02***	0.40***	-0.02***
dist_regional_50	-0.27***	-0.66***	-0.26***
dist_localbig_50	0.09***	0.71***	0.08***
dist_localsmall_50	-0.50***	-1.02***	-0.50***
AIC	3’047’602	103’234.5	2’935’663
BIC	3’047’838	103’407.5	2’935’898
Log Likelihood	-1’523’781	-51’597.2	-1’467’811
Deviance	1’275’973	28’363.47	1’246’256
Num. obs.	983’719	42’341	941’378

*P-value note: ***<0.001<**<0.01<*<0.05*

4. Discussion of the results and conclusions

Those empirical results (summarised in detail below) confirm the need to enrich the conceptual design of the entrepreneurial ecosystem by adding the impact of the population to a set of its determinants. The elasticity model that was used in our analysis is an operationalisation of the carrying capacity concept and supports the claim that population density can significantly stimulate firm births.

This paper adds a few novelties to business venturing literature. First, it supplements the somehow understudied social dimension of the entrepreneurial ecosystem and develops a coherent theory. Secondly, by focusing on the location of business ventures it develops a stream of localized entrepreneurial ecosystems. Third, through advanced modelling on individual data, it is not only a theoretical hypothesis but a study with real verification of the proposed theory.

Specific conclusions that can be drawn from the study presented can have profound implications for existing research:

- There exists a mechanism of attraction, which links the business activity to its root cause – population density. There is evidence that population density can be a significant factor in firms’ location.
- Economies of density appear hierarchically alongside economies of agglomeration. This is not a trade-off relation – economies of density and agglomeration co-exist and affect both 1st and 2nd line companies.
- The attraction mechanism works on different spatial levels (on the intra-urban micro-scale as well as on the intra-regional macro-scale), making it a universal and robust phenomenon.
- Business sectors differ in terms of location determinants (as in Ferreira et al. 2016). This means there is no one-fit-all theory linking the relationship between the economies of density and the economies of agglomeration. Model results show that some firms prefer more vibrant, urban places, while others prefer more secluded,

peripheral locations with lower population density, however, population density remains an important locational factor regardless of the sector.

- There is an evident and stable level of elasticity (ε) of business agglomeration with respect to the residential population. As it was shown, when local population density doubles, - local business agglomeration increases on average by ca. 76% (and even 79% for 1st line firms). The limits of space, congestion and concentration effects result in an elasticity of less than 1 ($\varepsilon<1$) eliminating infinite growth in the number of firms. In the most densely populated districts, more companies tend to locate in the suburbs (up to 25km from the main city), pushing the entrepreneurial potential stimulated by population growth towards peripheral areas. This highly spatial-related effect explains why direct suburbs and industrial rings are increasingly appealing for business while being less attractive for human settlements.
- Business benefits from two density-related phenomena: density externalities (understood as the number of inhabitants in a place) and urbanization externalities (understood as a high-density clustering). There is no overlap between the two– both effects are observable in explaining firms' location decisions.
- Depending on the sector, Marshallian or Jacobian agglomeration externalities can be observed. 2nd line firms often locate next to companies from the same sector. This Marshallian agglomeration effect can be explained by a path-dependent process which makes firms mimic entrepreneurs specialising in the same field, which also suggests homogeneity of (location-determining) factors for firms in a particular sector At the same time, almost all of the analysed sectors (except for 2nd line production firms) were attracted to areas with higher industrial diversity, so for both 1st line and most 2nd line firms the Jacobian type externalities matter.
- Quantitative methods proposed in this paper: mediation model, binary choice model and elasticity model are complementary approaches to track the analysed relations.

We believe that our results open a new path for research on entrepreneurship, business location and business demography. If the effects analysed in our study are not taken into account simultaneously, it may lead to biased coefficients in business choice models and even misleading conclusions in business location research. Focusing only on agglomeration economics (as in many previous studies) makes the analysis much more limited in terms of the effects related to the spatial location of firms. Confirming that population density is one of the root determinants of firm creation, its impact should not be overlooked in enterprise venturing studies.

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Appendices

Appendix 1: Economic sectors in REGON register

REGON register collects information on all existing firms in Poland. Data are publicly available[4]. The analysed dataset included all registered firms in 2012 in the NUTS1 Mazovian region (Poland). Sectors were coded with capital letters as follows (Tab.A1):

Table A1: Economic sectors used in analysis

Symb.	Abbreviation	Full name of sector	Number of obs.
A	Agri_forest	Agriculture, forestry, hunting and fishing	253'514
B	Mining	Mining and exploration	862
C	Indust_process	Industrial processing	58'824
D	Energy	Producing and supplying in electricity, gas, steam, hot water and air conditioning systems	1'944
E	Water_waste	Water supply; wastewater management, waste management and remediation activities	2'359
F	Constr	Construction	71'579
G	Trade	Wholesale and retail trade; repair of motor vehicles and motorcycles	191'442
H	Logistics	Transportation and storage	44'894
I	Hotels_catering	Activities related to accommodation and catering services	18'223
J	Inform_communic	Information and communication	38'822
K	Finance_insur	Financial and insurance activities	24'606
L	Real_estate_serv	Activities related to real estate services	32'056
M	Prof_scien_tech_serv	Professional, scientific and technical activities	97'762
N	Administr	Administration and support service activities	26'526
O	Security_publ_adm	Public administration and defense; compulsory social security	3'471
P	Educat	Education	26'746
Q	Healt_social_assist	Healthcare and social assistance	32'988
R	Culture_leasure	Activities related to arts, entertainment and recreation	11'847
S	Other_serv	Other service activities	45'101
T	Private_act	Private households hiring employees; households producing goods and providing services for their own needs,	5
U	Org_extraterit	Organisations and extraterritorial teams	144
TOTAL			**983'715**

This dataset includes also PKD7 (*Polska Klasyfikacja Dzialalności*) five digit code which describes the major declared activity of a given firm[5]. The analysed dataset included 646 individual PKD7 activities out of 654 available PKD7 codes.

PKD7 codes can be classified according to technology involvement. European classification (Eurostat, 2020[6]) is based on NACE codes[7]. Polish PKD7 classification is linked to NACE. Polish Agency for Enterprise Development (PARP, *Polska Agencja Rozwoju Przedsiębiorczości*) publishes PKD7 codes which are typical for business activities involving medium-high-technology, high-technology or high-tech knowledge-

[4] https://wyszukiwarkaregon.stat.gov.pl/appBIR/index.aspx
[5] https://www.biznes.gov.pl/pl/tabela-pkd
[6] https://ec.europa.eu/eurostat/cache/metadata/en/htec_esms.htm
[7] https://ec.europa.eu/competition/mergers/cases/index/nace_all.html

intensive services[8]. This classification was used to distinguish high-tech firms. Data was recoded using Polish PKD classification being equivalent to NACE classification.

REGON dataset includes also information on employment size. Variable *Gr_empl* reports five employment classes: class 1 for up to 9 persons, class 2 for 10-49 persons, class 3 for 50-249 persons, class 4 for 250-1000 persons and class 5 for more than 1000 persons. Variable *Empl* gives the approximate mid-value of group: 5 for class 1, 30 for class 2, 150 for class 3, 600 for class 6 and 1500 for class 5.

[8] https://www.parp.gov.pl/storage/grants/documents/41/Eurostat_pkd_333.pdf

Appendix 2: Location of regional cities and DBSCAN clusters

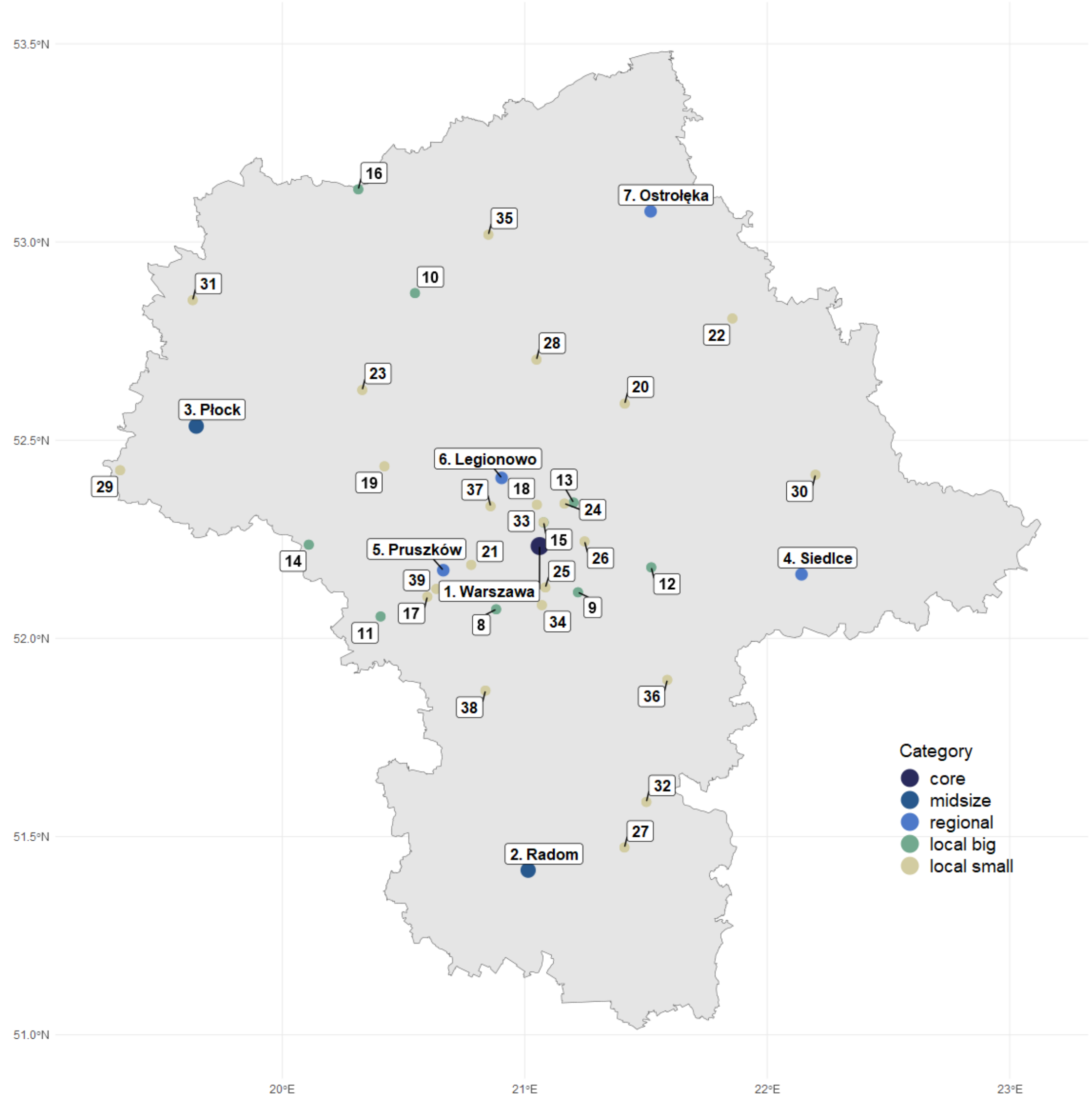
Relative location to cities was derived using real locations of 39 cities in the Mazovian NUTS1 region (*województwo mazowieckie*), Poland (Tab.A2). They were grouped into 5 population size: core 1mln+ (1 city, Warsaw), midsize 100K+ (2 cities, Radom and Płock), regional 50K+ (4 cities, Siedlce, Pruszków, Legionowo, Ostrołęka), local big 30K+ (9 cities), local small 15-30K (23 cities). Each firm was checked if it is located in a radius of 10, 25 and 50 km from the centre of each city. Finally, instead of distances to particular cities, the dataset included 15 dummy variables: *dist_core_10, dist_core_25, dist_core_50, dist_midsize_10, dist_midsize_25, dist_midsize_50, dist_regional_10, dist_regional_25, dist_regional_50, dist_localbig_10, dist_localbig_25, dist_localbig_50, dist_localsmall_10, dist_localsmall_25, dist_localsmall_50*.

Table A2: Cities located in NUTS1 Mazovian region included in the analysis

ID	city	Population (persons)	area (km2)	Density	longitude	latitude
1	Warszawa	1'729'119	517.9	3339	21.06119	52.23294
2	Radom	217'834	111.8	1948	21.01389	51.4152
3	Płock	122'572	88.06	1392	19.64501	52.53549
4	Siedlce	76'585	31.87	2403	22.14144	52.16167
5	Pruszków	59'796	19.19	3116	20.66378	52.17202
6	Legionowo	54'246	13.6	3989	20.90458	52.40488
7	Ostrołęka	52'792	29	1820	21.51907	53.07745
8	Piaseczno	45'270	16.22	2791	20.88196	52.07337
9	Otwock	45'073	47.33	952	21.21945	52.11633
10	Ciechanów	44'673	32.84	1360	20.54715	52.87107
11	Żyrardów	41'056	14.35	2861	20.40536	52.05526
12	Mińsk Mazowiecki	40'028	13.12	3051	21.52169	52.17917
13	Wołomin	37'418	17.24	2170	21.20118	52.34264
14	Sochaczew	37'333	26.13	1429	20.1092	52.23635
15	Ząbki	32'376	11.13	2909	21.07776	52.29252
16	Mława	30'893	35.5	870	20.31333	53.13291
17	Grodzisk Mazowiecki	29'988	13.19	2274	20.59806	52.10513
18	Marki	29'395	26.03	1129	21.04995	52.33716
19	Nowy Dwór Mazowiecki	28'361	28.27	1003	20.42122	52.43413
20	Wyszków	27'205	20.78	1305	21.41228	52.5923
21	Piastów	22'862	5.76	3969	20.77894	52.18553
22	Ostrów Mazowiecka	22'770	22.09	1031	21.85631	52.80713
23	Płońsk	22'435	11.6	1934	20.32957	52.62666
24	Kobyłka	21132	19.64	1076	21.16377	52.34008
25	Józefów	20013	23.92	837	21.08495	52.12857
26	Sulejówek	19'385	19.31	1004	21.24671	52.24465
27	Pionki	19'286	18.34	1052	21.41105	51.47252
28	Pułtusk	19'229	23	836	21.04836	52.70321
29	Gostynin	19'026	32.4	587	19.33104	52.42421
30	Sokołów Podlaski	18'730	17.5	1070	22.19902	52.41285
31	Sierpc	18'468	18.6	993	19.63032	52.85337

32	Kozienice	18’150	10.45	1737	21.50136	51.58732
33	Zielonka	17’434	79.48	219	21.07706	52.293
34	Konstancin-Jeziorna	17’371	17.74	979	21.07058	52.08389
35	Przasnysz	17’337	25.16	689	20.84989	53.01855
36	Garwolin	17’160	22.08	777	21.58779	51.89536
37	Łomianki	16’632	8.4	1980	20.85897	52.33345
38	Grójec	16’430	8.57	1917	20.83774	51.86819
39	Milanówek	16’427	13.52	1215	20.63528	52.12485

Figure A1: Urban settlement in Mazovian region

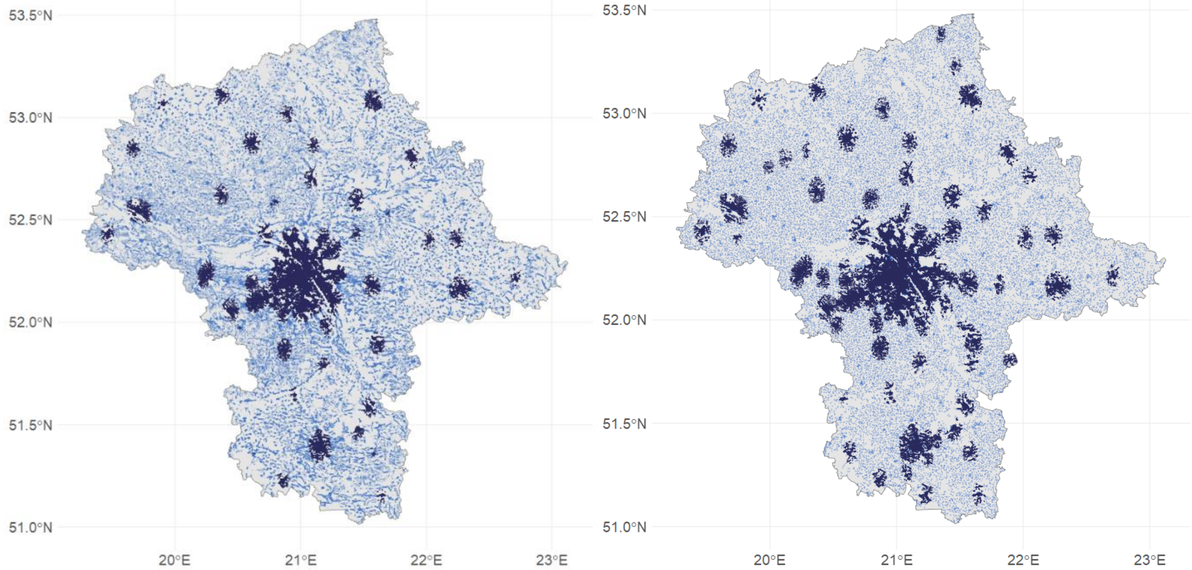


Source: Own Work

DBSCAN algorithm is density-based clustering. It divides geo-localised points into core points (located in high-density areas) and noise (located in low-density areas) – there is no pre-assumption on a number of clusters. The clustering mechanism uses for each point a circle of specified radius and checks how many points are in this circle – if the number of points exceeds the assumed threshold, the given location is classified as core, if not – as noise. In this study, we took a radius of 0.03° what is equivalent to ca.3.3

km and minPts=75 firms / 500 persons. The selection of these parameters involved a deeper analysis of the stability of the solution. An alternative way to DBSCAN could be setting a threshold for the local population or business density and delineating locations belonging to high-density clusters. These methods can be applied to regional or grid data, while their main disadvantages remain all problems linked to data aggregation, including MAUP (Modified Areal Unit Problem).

Figure A2: Spatial data used in analysis: a) business location with DBSCAN high-density clusters, b) population location with DBSCAN high-density clusters

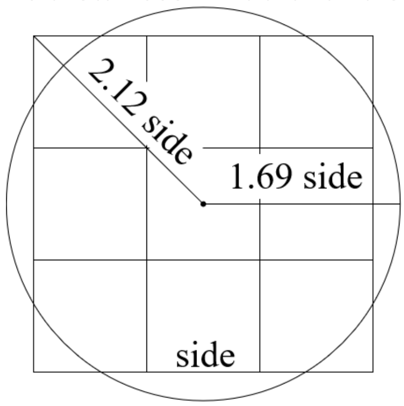


Source: Own Work

Appendix 3: Spatial granulation – justification of circles and grid-to-point disaggregation

Using a circle as a bounding box for the spatial neighbourhood is well justified with other spatial methods also based on circles. One should mention here: i) DBSCAN, which is testing how many points are in a specified radius to detect high-density areas (Ester et al., 1996), ii) Geographically Weighted Regression (GWR) which performs local regressions in optimised bandwidth (radius) (Brunsdon et al., 1998), iii) Ripley's K which outputs the agglomeration function by counting points in increasing radii, iv) SPAG which calculates area covered by overlapping circles representing points to measure agglomeration synthetically (Kopczewska et al., 2019), v) Kuldorf's spatial scan statistics which compares within a moving ring the probability of being the case given populations at risk inside and outside the ring (Kulldorff, 1997). This popularity of circular neighbourhoods results from their natural properties. Let's compare the circle with the commonly used grid neighbourhood in a queen form (all eight surrounding boxes neighbouring with a selected one). Assuming that the areas of 9 squares ($9*side^2$) and a single circle ($\pi*r^2$) are equal, the radius of the circle equals 1.69 of the single square's side (Fig.A3). This matters for the relation of the maximum distance between the core and the most extreme point. In the radial setting, maximum distance equals r (1.69*side), while in the squared setting, it is $3*side*\sqrt{2}\ /2=2.12*side$, what makes maximum distance in squared setting longer by 25%. Thus, with the same "catchment" capacity in a squared setting (the same area of analysis), the distances are longer which makes the potential distance-decaying pattern more intensively represented. This suggests that the power of spatial interactions is more diversified in squared data, thus, modelling may be less sensitive. This leads to the conclusion that, if possible, a radial approach should be used to deal with more coherent data.

Figure A3: Visualisation of the distances in radial and squared approaches

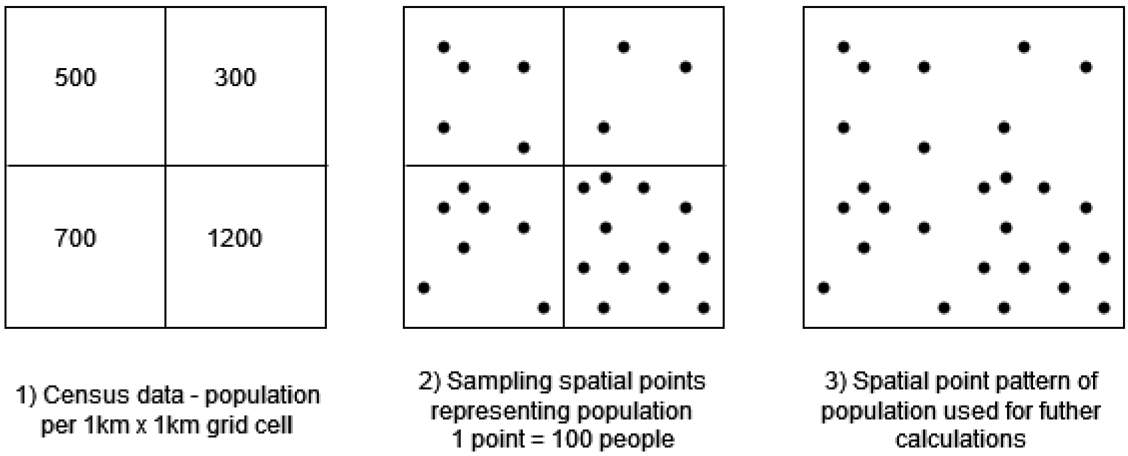


Source: Own Work

Population data, available as grid data, were disaggregated into point data. The sampling procedure assumed drawing for each grid cell (1 km x 1 km) from random spatial distribution the equivalent number of points (1 point = 100 persons) (Fig.A4). Those random points were geo-localised (longitude/latitude) and treated as point pattern. This newly created dataset was used in a few computations:

- To detect DBSCAN clusters with a high-density of population
- To find local agglomeration of population, by counting the number of inhabitants in a radius of 500 m from a given firm

Figure A4: Creating population point pattern from grid census data



Source: Own Work

Appendix 4: Details of probit, random forest and multidimensional scaling

Selection of binary-choice model over e.g. count models has a methodological justification. Many studies (e.g. Carpenter et al., 2021) use count data as dependent variable to express the popularity of given surrounding (box, circle) for business location. This approach changes the research question – from location decision to determinants of clusters; and estimation method – from binary models to count models.

Probit model is a classical econometric solution for the binary dependent variable. It models the probability of success – that an event occurs. Probit beta coefficients show an impact of unitary change of the explanatory variable (*ceteris paribus*) on the probability that an event appears – an increase in variables with positive coefficients increases the probability of the event. Model quality can be measured in absolute terms with McFadden R^2 (interpreted as a classical one, the higher the better, value close to 1 for a perfect fit) and relatively, between the models, with AIC (Akaike Information Criterion) (the smaller the better, AIC depends on a number of observations). Even if probit is less restrictive (than e.g. logit) regarding the normality of variables' distributions, it has the disadvantage of being a linear combination of variables. Therefore the cross-check is being done with a supervised machine learning model – random forest, which is more flexible in the case of non-linear relationships and models inhomogeneities much better.

Random forest model is a well-established supervised machine learning method, being a mixture of decision tree and bootstrap. It can also be applied in spatial modelling (Kopczewska, 2022). It samples observations and variables in each iteration (tree) to find the best cut-off points of the explanatory variables in order to get coherent groups indicated by binary levels of the dependent variable. The aggregation of bootstrap results from random forest ensemble is by majority voting (bagging). The output of random forest is variable importance – information on how much the variable matters in the model. Two criteria may be used: misclassification rate, which measures the ratio of miss-classified predictions when the variable of interest has been shuffled (randomised); or decrease in node purity, which measures how much MSE (*mean square error*) is reduced when a variable of interest becomes a node split of the tree. Both criteria are measured across all trees: misclassification as a ratio of wrong predictions, and node purity as a total reduction of MSE. In both cases, the higher the indicator, the more important the variable is in predictions. For multiple models, variable importance can be reported as box plots (instead of bar plots). The quality of the random forest is typically assessed with confusion matrix (CM) and out-of-bag (OOB) prior error rate. A confusion matrix (CF) is a two-dimensional contingency table, which counts observed and predicted labels (0 and 1). Classification errors in CM (for label=0 and label=1) measure the fit misclassification ratio:

$$err_{obs=0} = \frac{count_{obs=0,fit=1}}{count_{obs=0,fit=0} + count_{obs=0,fit=1}} \tag{A.1}$$

and

$$err_{obs=1} = \frac{count_{obs=1,fit=0}}{count_{obs=1,fit=1} + count_{obs=1,fit=0}} \tag{A.2}$$

The lower the error the better the predictions from the model. Out-of-bag data are observations not included in the estimation of a given tree (due to bootstrap sampling), which are used for predictions of a given tree. OOB prior error rate is the misclassification rate of OOB data (it plays the role of randomised cross-validation) – the smaller the error the better.

Multidimensional scaling – projects k-dimensional data into n-dimensional points (n<k), mostly in form of 2D plot. It uses Kruskal scaling (1964) to calculate the Stress function expressed as:

$$STRESS = \sqrt{\frac{\sum\left(D_{ij} - d_{ij}\right)^2}{\sum\left(D_{ij}\right)^2}} \tag{A.3}$$

where *i and j* are iteration operators $i, j \epsilon [1, k]$ (to iterate by all pairs of variables), D_{ij} are the distances in original *k* dimensions between variables *i* and *j*, and d_{ij} are distances in a new (2D) dimensions. Outcomes are 2D locations of points which illustrate the original distances best.

It can deal with any type of data, also with dummy variables by using appropriate distance metrics. One of the solutions is to use Gower distance which can cope with mixed data types – continuous and binary variables together. The optimisation algorithm finds locations of points in 2D space to minimise the Stress function, which expresses the difference between original and new distance relations. Its main advantages are flexibility due to types of data, joint (rather than pair by pair) analysis of variables, and efficient visualisation.

Appendix 5: Empirical results of data analysis

All computations were made in R software. For processing of geo-localised data we used {sp}, {rgeos}, {spdep}, {spatstat}, {rgdal} and {sf} packages. Density clusters and neighbourhood locations were found with {dbscan} and {nabor}. Multidimensional scaling was made with {smacof}. The modelling used {abdrf} for the random forest, {lavaan} and {mediation} for the mediation model and {ivreg} for the mediation model with instrumental variables. For synthetic outputting, we applied {stargazer} and {texreg}. Parallel computations were executed with {doParallel}.

Tab.A3 presents the descriptive statistics of the analysed dataset – for pooled data and in the division for 1st line and 2nd line firms. For pooled data, we present mean, standard deviation, and extreme values (minimum and maximum). For data in groups we present average values only – extreme values were similar, while the variance in subgroups was quite narrow.

Table A3: Descriptive statistics of the dataset

		All firms					1st line	2nd line firms			
		pooled data						agricul-ture	produ-ction	constru-ction	service
	no of obs →	983 719					42 341	253 514	55 969	71 579	560 312
	Statistic	mean	st.dev	min	max		mean	mean	mean	mean	mean
neighbourhood	locPdens	82.88	90.12	0.00	376.00		126.14	15.05	81.21	86.64	110.42
	locAggAgri	86.04	146.41	0.00	1.91		78.89	86.63	78.74	84.78	87.20
	locAggProd	105.62	133.19	0.00	865.00		160.21	19.83	110.66	110.23	139.22
	locAggConstr	118.67	145.54	0.00	876.00		178.85	23.85	119.95	125.54	156.02
	locAggServ	1 297	1 761	0.00	11 537		2 055	202.46	1 268	1 310	1 736
	locBIG	13.83	24.69	0.00	174.00		21.85	1.86	13.68	13.76	18.67
	locHH	0.00	0.0001	0.00	0.07		0.00	0.00	0.00	0.00	0.00
	locLQ	1.54	0.90	0.00	5.00		1.59	2.16	1.31	1.10	1.34
	COREfirms	0.69	0.46	0.00	1.00		0.94	0.21	0.78	0.77	0.87
	COREpopul	0.75	0.43	0.00	1.00		0.97	0.31	0.85	0.84	0.91
location	dist_core_10	0.33	0.47	0.00	1.00		0.57	0.03	0.33	0.34	0.45
	dist_midsize_10	0.03	0.17	0.00	1.00		0.02	0.02	0.04	0.04	0.03
	dist_regional_10	0.08	0.27	0.00	1.00		0.08	0.05	0.10	0.09	0.09
	dist_localbig_10	0.34	0.47	0.00	1.00		0.45	0.10	0.38	0.37	0.43
	dist_localsmall_10	0.54	0.50	0.00	1.00		0.78	0.21	0.59	0.59	0.67
	dist_core_25	0.53	0.50	0.00	1.00		0.84	0.09	0.59	0.57	0.69
	dist_midsize_25	0.09	0.28	0.00	1.00		0.04	0.11	0.11	0.09	0.07
	dist_regional_25	0.53	0.50	0.00	1.00		0.72	0.22	0.56	0.56	0.65
	dist_localbig_25	0.66	0.47	0.00	1.00		0.91	0.31	0.71	0.71	0.79
	dist_localsmall_25	0.84	0.37	0.00	1.00		0.96	0.66	0.87	0.87	0.90
	dist_core_50	0.64	0.48	0.00	1.00		0.90	0.27	0.70	0.70	0.78
	dist_midsize_50	0.16	0.37	0.00	1.00		0.05	0.29	0.16	0.15	0.11
	dist_regional_50	0.80	0.40	0.00	1.00		0.94	0.60	0.81	0.82	0.87
	dist_localbig_50	0.82	0.39	0.00	1.00		0.95	0.63	0.83	0.84	0.88
	dist_localsmall_50	0.99	0.10	0.00	1.00		1.00	0.97	0.99	0.99	1.00

Tab.A4 gives the details of the quality assessment for the Random Forest model. It reports the confusion matrix (CM) for 5 models. Classification errors were calculated as described in Appendix 4.

Table A4: Quality metrics of random forest models based on confusion matrix (CM)

	Model	Label	CM 0 observed	CM 1 observed	Classification error	Out-of-bag (OOB) prior error rate
1st line	1st line firms	CM_0_predicted	940’597	781	0.0008	0.0371
		CM_1_predicted	35’732	6’609	0.8439	
2nd line	Agriculture	CM_0_predicted	679’408	8’456	0.0122	0.0163
		CM_1_predicted	6’920	246’594	0.0272	
	Production	CM_0_predicted	884’935	474	0.0005	0.0367
		CM_1_predicted	34’065	21’904	0.6086	
	Construction	CM_0_predicted	869’212	587	0.0006	0.0425
		CM_1_predicted	39’453	32’126	0.5511	
	Service	CM_0_predicted	321’286	59’780	0.1568	0.0800
		CM_1_predicted	15’569	544’743	0.0277	

Appendix 6: Endogeneity and causality – mediation model with instrumental variables

As stated in the text, the mediation model was applied to examine the hierarchical impact of population density on business agglomeration. The direct impact on 1st line firms and the indirect impact via 2nd line firms were analysed, and it was concluded that this mechanism indeed exists and mediation occurs. However, one ambiguity remained regarding the potential backward direct relation in which the 1st line firms attract the population.

To address this issue, a two-step methodology was employed (Fig.A5). The first stage tests the potential endogeneity between firms and the population with a standard instrumental variable (IV) approach – this is to asses any potential feedback loops. It examines the unmediated relations in two separate IV equations and evaluates the endogeneity on both the 1st line and 2nd line firms. This procedure was carried out with a standard IV methodology using 2SLS estimation. The tested relations are illustrated in eq.1a and eq.2a and were augmented with the IV relation (eq.1b and eq.2b).

$$1stLine_i = \alpha_0 + a \cdot population_i + \varepsilon_i \tag{1a}$$

$$population_i = d_0 + d \cdot instrument_i + \varepsilon_i \tag{1b}$$

$$2ndLine_i = b_0 + b \cdot population_i + \varepsilon_i \tag{2a}$$

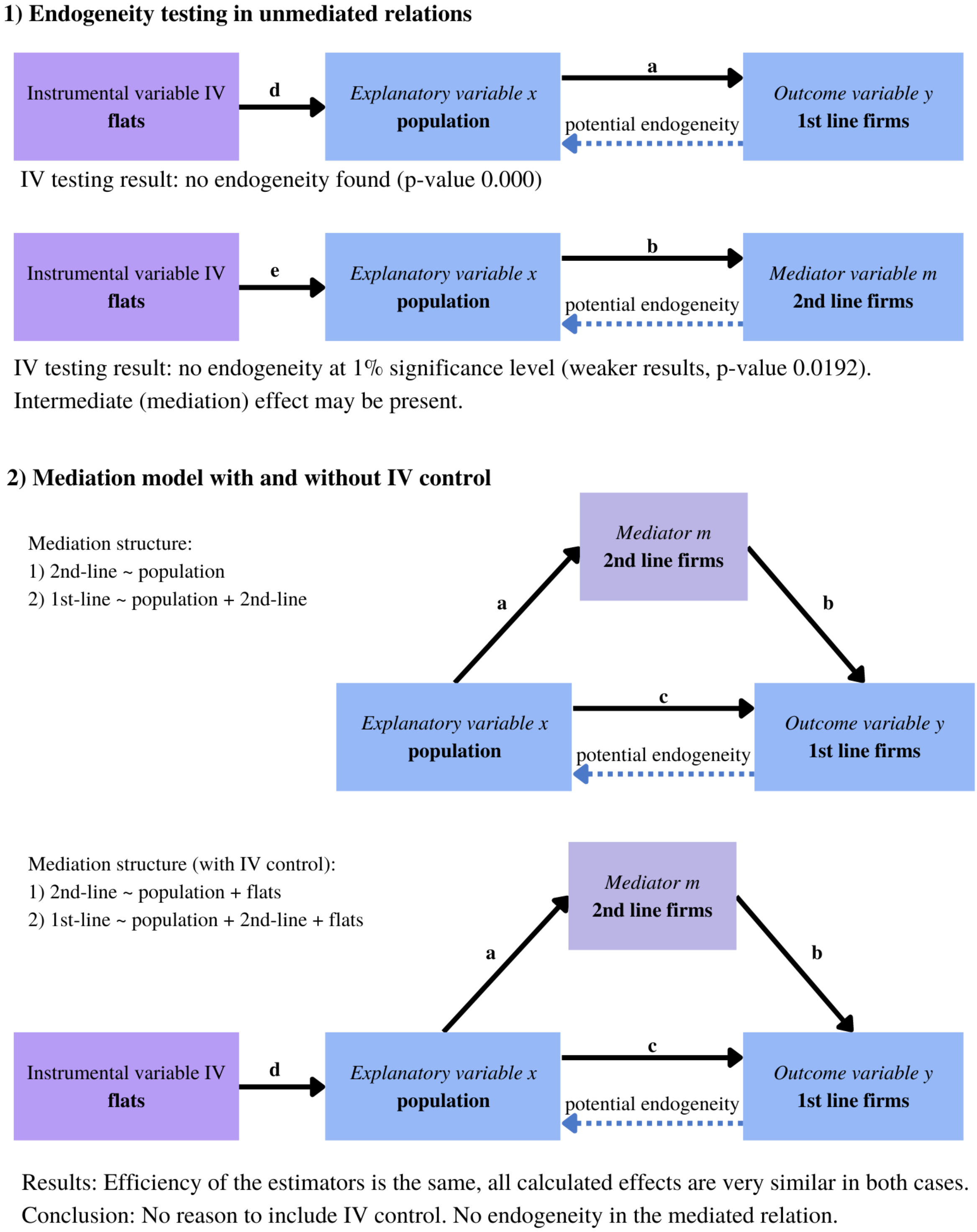
$$population_i = e_0 + e \cdot instrument_i + \varepsilon_i \tag{2b}$$

The second stage links the mediation model and instrumental variables (IV), following the method by Dippel et al. (2020) to validate the robustness of the findings. This entailed testing the model with (eq.3) and without additional IV control (eq.4). This approach requires the inclusion of an instrumental variable that is correlated with the explanatory variable but not with the dependent variable. In this case, a historical IV was used – information from the past that holds the mentioned relations with variables in the model. A comprehensive discussion on the selection of instruments can be found in Puga (2010).

$$\begin{gathered} 2ndLine_i = \alpha_0 + a \cdot population_i + d \cdot instrument_i + \varepsilon_i \\ 1stLine_i = a_1 + c \cdot population_i + b \cdot 2ndLine_i + d \cdot instrument_i + \varepsilon_i \end{gathered} \tag{3}$$

$$\begin{gathered} 2ndLine_i = \alpha_0 + a \cdot population_i + \varepsilon_i \\ 1stLine_i = a_1 + c \cdot population_i + b \cdot 2ndLine_i + \varepsilon_i \end{gathered} \tag{4}$$

Figure A5. Structure of the endogeneity testing – IV regression of population and firms, and two mediation models – with and without the additional control from the instrumental variable



Source: Own work

An issue of spatial granulation was encountered in the analysis, as the main study was conducted on individual geo-located point data. However, in the past, such detailed statistical information was not collected, and the lowest aggregation level is a municipality. In Poland, LAU2 is the lowest aggregation with available historical data and the only level that is comparable over time[9]. Therefore, the IV mediation regression, which includes historical data, was conducted on LAU2 rather than on individual points. This approach also allows for the generalisation of the studied mechanism, as it covers the whole territory of the country (ca. 2500 LAU2 units).

The instrument chosen to test for backward causality is the number of flats available in 1988 at the municipality level. This variable is well-connected to the current population and has no relation to a current business location. The high correlation (0.98) between the flats available in the 1988 and 2021 censuses and the population registered in the census ensures that the lodgings build in the past set the trends for the modern housing market in Poland. Moreover, the 1988 data comes from the last national census organized under the communist system, making the housing system unrelated to the later introduced free market while still ensuring data reliability (the raw calculations from the 1988 census were reviewed again in 1998 after the independent Main Statistical Office was reinstated in Poland). The remaining variables used to test endogeneity follow the structure of the main study, with the difference that here the data is aggregated at the municipality level and compares the current values with the historical instrument. The 1st line firms variable is measured as the number of companies operating within knowledge-intensive sectors at LAU2 in 2021. The 2nd line firms variable is, respectively, measured as the number of companies specialising in retail, catering and less-productive services (2nd line firms) in 2021. Population data is taken from the recent census (2021) and measured at the LAU2 level. All variables used for endogeneity testing were transformed using the logarithmic function.

Two IV regressions were performed to statistically test whether there is endogeneity between 1st or 2nd line firms and the population (Fig.A5). The results showed a positive and significant relationship between both types of firms and the population, with the number of flats serving as a strong instrument, confirming the validity of the chosen IV (Tab.A5). However, the endogeneity testing was not straightforward, and the interpretation is unclear for the 1st line firms. While the Wu-Hausman test did not reject the null hypothesis of no endogeneity between 2nd line firms and the population density, a higher significance level might suggest the presence of some hidden endogeneity between the population and 1st line firms. In other words, there is no endogeneity in the relationship between population and 2nd line firms, but in the case of 1st line business, the hypothesis can be rejected at a higher level of statistical significance than 1%. This result does not contradict the logic presented in the main paper and suggests that there might be a hidden factor (or a mediator) that should be considered in the relationship between innovative companies and population density – the 2nd line business.

Table A5. Instrumental variable regressions results

1st-line firms			
Coefficients	**Estimate**	**Std. Error**	**p-value**
(Intercept)	-8.37302	0.10494	0.0000

[9]In 1999 Poland undergone deep territorial reform as the part of European Union accession – this harmonization of territorial division of the country was to adjust the statistical reporting to rules set by EUROSTAT. LAU2 unit (municipalities, *pl gmina*) were kept fixed as they were, while other upper levels of territorial division (LAU1, NUTS3, NUTS2, NUTS1) were newly introduced. Number of regions decreased from 49 to 16.

log(population)	1.50689	0.01225	0.0000
Tests	**Df1 / Df2**	**Statistic**	**p-value**
Weak instruments	1 / 2374	16009.727	0.0000
Wu-Hausman	1 / 2473	5.491	0.0192
2^{nd}-line firms			
Coefficients	**Estimate**	**Std. Error**	**p-value**
(Intercept)	-4.689873	0.072406	0.0000
log(population)	1.192730	0.008454	0.0000
Tests	**Df1 Df2**	**Statistic**	**p-value**
Weak instruments	1 2374	16009.727	0.0000
Wu-Hausman	1 2473	0.007	0.9360

The mediation analysis was repeated on the aggregated data to verify if the conclusions obtained in the main study can be generalized for the whole country data. The mediation model was estimated in two versions – with and without the IV control – and the efficiency of the estimators was compared to show if the additional IV control is necessary (Fig.A5). The results of the standard (non-IV) mediation model run of LAU2 data show that all tested effects are highly significant (Tab.A6 – estimates without IV). Strong and positive direct and total effects indicate the causal impact of population density on the emergence of 1^{st} line firms. This impact is mediated by the presence of 2^{nd} line firms, which account for 72% of the total effect. The relationship between population density and 1^{st} line firms is again significant and positive. The results obtained with the mediation model on LAU2 data support our claim that population density is a strong predictor for the emergence of innovative businesses, both directly and via the mediation effect of less-productive service and retail companies. These results are highly comparable to the main study conducted on individual geo-located point data presented in the paper.

Table A6. Mediation models on aggregated data – with and without IV control

Effect	Estimate without IV	Estimate with IV	% difference
ACME	1.09336*** [1.04194; 1.15218]	1.09526*** [1.02509; 1.16111]	- 0.17%
ADE	0.42444*** [0.36282; 0.47546]	0.49212*** [0.42293; 0.57061]	- 15.95%
Total Effect	1.51779*** [1.49457; 1.53821]	1.58738*** [1.52996; 1.63923]	- 4.58%
Proportion mediated	0.72116*** [0.68718; 0.76006]	0.69019*** [0.64575; 0.73052]	4.29%
Significance codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1			

A comparison of the results from both mediation models (with and without IV) was conducted to ensure that the conclusions remain the same regardless of the inclusion of the instrumental variable. The estimated effects are presented in Tab.A6, which shows that the coefficients remained almost the same in both models. The estimation did not gain efficiency when the additional control was included, which proves that there is no significant endogeneity in the hierarchical relation between population density and the number of 1st line firms, mediated by the 2nd line firms.

Table A7. Sensitivity analysis of both mediation models

Sensitivity score	Model without IV	Model with IV
Rho at which ACME=0	0.6000	0.6000
Rho at which ADE = 0	-0.3000	-0.3000

Additional support to this claim is provided by the results of the sensitivity analysis (Imai et al. 2010) for the mediation models, which were run for both IV and non-IV models. The sensitivity parameters are the same in both cases, with *rho* for the Average Causal Mediation Effect (ACME) being 0.6, and for the Average Direct Effect (ADE) rho equals -0.3 (Tab.A7). Both show considerable parameter switches necessary to overturn the conclusions shown in the model. This observation, together with the narrow confidence intervals for the estimated effects, gives us reason to trust the reasoning presented here.

To explain the emergence of 1st line firms in a given area, one should take into consideration the local population density and the mediation effect of the 2nd line firms. The emergence of 1st line firms is caused by the local population density, and the effect is mediated by the intermediate variable, which is the number of 2nd line firms operating nearby. If the mediation effect is taken into consideration, the population effect is not endogenous to the model. These two variables are enough to assert the direction of impact – local population density is the cause of innovative companies' location.

References to the appendix

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Appendix 7: Alternative classifications of firms

Literature offers few classifications of firms, however, they do not fit the presented concept due to no reference to technology or spatial location. One can mention few:

a) basic and non-basic firms (basic firms making their incomes with external customers, while non-basic earning money on residents) – it does not deal with technology, as the criterion is source of income
b) tradable and non-tradable jobs/sectors (non-tradable jobs generate service consumed in place of creation, while tradable jobs generate products that can be shipped) – it does not deal with technology

c) high-tech knowledge-intensive firms and low-tech low-knowledge-intensive firms – even if it deals with technology, classification does not account for daily needs
d) "Schumpeterian firms" (disruptive firms) and "Kirznerian firms" (equilibrating firms) (Langlois, 2002) – requires not straightforward analysis of opportunities on disruption, information needs, innovativeness, commonness, and discoveries (de Jong & Marsili, 2015)
e) "high-end" and "low-end" firms – no middle firms